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AGI JOBS × OPENAI GYM × SUCCESSOR Ω

Formal Specification

Typed AGI Job contracts, Mission Gym semantics, proof-preserving composition, machine-readable records and bounded authority execution

WORK, EXPERIENCE AND AUTHORITY REMAIN SEPARATE TYPES

This volume defines the implementable constitution : how jobs compose evidence without composing authority ; how the Gym produces attributable experience ; how exact releases are frozen ; how proof is signed ; and how Successor Omega operates through machine-enforced scope, rollback and expiry.

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This companion is part of the same versioned publication architecture. The flagship paper presents the central argument and essential results. This volume preserves the deeper specification or evidence needed for implementation, reproduction and adversarial review. All state claims remain bounded by the same rule : authority never exceeds fresh proof.

Canonical progression

Successor Manifold ⟶ Mission Advantage Gradient ⟶ SEIZE Underwriting ⟶ Bounded AGI Jobs ⟶ Formation Gym ⟶ Exact Release Freeze ⟶ Fresh Proof ⟶ Specialist ASI ⟶ Successor Ω.

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PART I

Formal Specification

The complete institution is executable only when work, evidence and authority remain separately typed

Typed work, environment and authority objects, including exact semantics, composition rules, invalidation, proof and execution.

SUCCESSOR MANIFOLD → GRADIENT → SEIZE → BOUNDED JOBS → SPECIALIST ASI → SUCCESSOR Ω

1 The Inner Mission Triad : Gym, Specialist ASI and Successor Ω

1.1 The triad

Navigator Ω and the Verified Succession Institution form the outer sensing, underwriting and renewal control plane. Inside each constituted mission, the terms Mission Gym, Specialist ASI and Mission-Sovereign Successor Ω refer to different objects and answer different questions.

Table 1. The Gym–ASI–Successor triad.

Object	What it is	Question answered
Mission Gym	A controlled, versioned representation of the mission's state, role-specific observations, actions, transitions, outcomes, constraints, scenario distributions and independent verification.	Can this architecture actually perform the mission, fail safely and transfer beyond development cases?
Specialist ASI	The complete candidate intelligence institution that proves mission-specific superiority over the incumbent and the best credible general, specialist, human, software or hybrid alternative.	Which complete intelligence institution wins the mission under fresh proof?
Mission-Sovereign Successor	The proven Specialist ASI after separate accountable admission, with identity, evidence, Chronicle, monitoring, rollback and a scoped Authority Envelope.	What authority should the institution grant this proven capability, for which version, scope, duration and conditions?

The three objects form a sequence but not a collapse :

$$\boxed{\text{Mission Gym} \neq \text{Specialist ASI} \neq \text{Mission-Sovereign Successor } \Omega.} \tag{1}$$

1.2 The Gym is not the intelligence

The Gym is the mission's controlled world. It contains the rules, data interfaces, constraints, scenario generator, hidden truth where available, and verification logic. A candidate can interact with it, but the Gym does not itself decide what to do. It may contain simulations, historical replay, live shadow interfaces, digital twins, deterministic checks and strategic counterparties. It is owned and governed by the institution or an independent proof custodian.

A Gym can be valuable even when no candidate qualifies as Specialist ASI. It can demonstrate that the incumbent remains superior, that a proposed reward is exploitable, that a provider's performance does not transfer, or that the mission is not sufficiently measurable. A clean rejection is an output of the Gym, not a failure of the Gym.

1.3 Specialist ASI is not one policy

A policy $\pi(a \mid h)$ maps a history or observation to an action distribution. A Specialist ASI can include many policies and non-policy components. Its intelligence may arise from :

- a portfolio of general and specialist models ;
- explicit world models and causal representations ;
- retrieval, tools, programs, search and simulators ;
- a producer, critic, verifier and governor ;
- human experts and accountable professional authorities ;
- a shared evidence graph and uncertainty model ;
- workflow integration and institutional memory.

The complete system may be superhuman on the constituted mission even if no individual component is. Conversely, a component can be superhuman on a subtask while the complete institution remains inferior because coordination, evidence, cost, reliability or authority fails.

Definition 1.1 (Mission-Dominant Specialist ASI). A complete mission-specific intelligence institution that exceeds the incumbent and the best credible general, specialist, human, software or hybrid alternative by a preregistered positive margin on fresh, protected and representative work, after quality, critical-error risk, complete economics, human burden, rights, sovereignty, resilience and governance are counted.

This definition follows the system-level and mission-bounded claim discipline established in the foundational GoalOS paper (Boucher, 2026). It does not imply universal intelligence.

1.4 Successor Ω is not unlimited autonomy

The symbol Ω denotes the complete institutional state in which a proven mission capability has been constituted, admitted and placed under continuous reproof. It is not legal personhood, self-ownership or unlimited sovereignty. The successor remains subordinate to accountable human and institutional principals.

Definition 1.2 (Mission-Sovereign Successor). A Mission-Dominant Specialist ASI separately admitted by accountable authority under a scoped, versioned, monitored, expiring and revocable Authority Envelope, with proof-carrying action, Chronicle memory, degraded modes, rollback and requalification.

The progression is

$$\begin{aligned}\text{Candidate Successor} &\longrightarrow \text{Verified Specialist Intelligence} \longrightarrow \text{Mission-Dominant Specialist ASI} \\ &\longrightarrow \text{Mission-Sovereign Successor.}\end{aligned} \tag{2}$$

1.5 The Successor Complex

The strongest object of analysis is the **Successor Complex** :

$$\mathfrak{X}_m = \langle \mathscr{G}_m, \gg, \mathscr{P}, \mathit{Chronicle}, \mathscr{A} \rangle, \tag{3}$$

where $\mathscr{G}_m$ is the sovereign mission environment, $\gg$ the complete candidate or admitted successor institution, $\mathscr{P}$ the independent protected evaluation plane, *Chronicle* the admitted evidence and memory state, and $\mathscr{A}$ the current Authority Envelope.

This representation prevents a common category error : treating model performance as if it already contained environment validity, proof integrity, rights, operations, memory and authority.

Triad law

The Gym makes the mission executable. Specialist ASI establishes which complete intelligence institution wins. Successor Ω determines how the proven winner may enter reality. Chronicle determines what survives long enough to shape the next succession.

1.6 The canonical progression

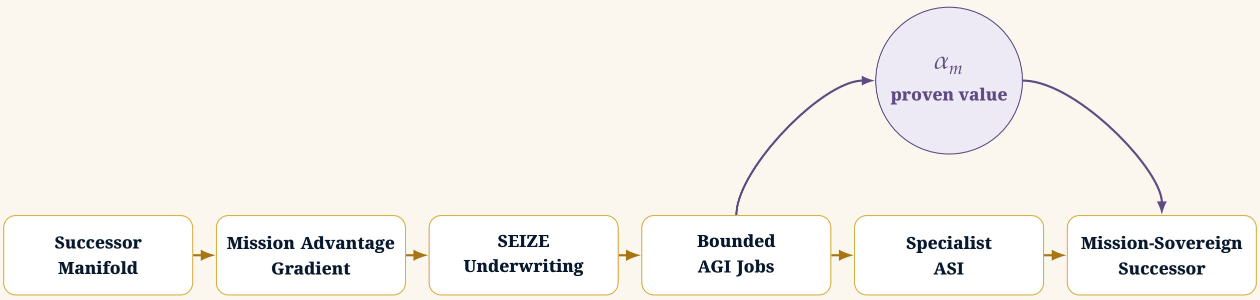

Figure 1. The canonical progression. Alpha is the residual value field produced by proof; it is not an independent chronological stage.

2 Successor Ω as the Complete Intelligence Institution

2.1 A system-level definition

Definition 2.1 (Successor Ω). A Successor Ω is an exact, mission-specific, versioned and accountable intelligence institution that has (i) passed fresh comparative proof against the incumbent and the best credible alternative, (ii) been separately admitted by an authorized principal, and (iii) received only the scoped, monitored, expiring and revocable authority supported by current evidence.

A Successor Ω is larger than the policy that selects actions. Let

$$\gg^{\Omega}_{m} = (M, D, H, T, V, W, R, I, E, C, A, P), \tag{4}$$

where M is the model and policy portfolio; D the rights-cleared data; H the harness and orchestration; T tools, retrieval, deterministic components and planners; V the independent verifier; W the workflow and AGI Job graph; R rights, security, resilience and rollback; I integration; E the evidence record; C Chronicle; A the Authority Envelope; and P the accountable human or institutional principal.

A model can be changed without changing the mission institution if the mission-level observables, evidence chain and authority invariants remain within preregistered tolerance. Conversely, changing a small prompt can create a new release if the change alters action selection, evidence, cost, rights or failure modes. Exact release identity is therefore institutional rather than cosmetic.

2.2 The Omega state

The symbol Ω denotes constitutional closure for one mission and one release, not universal completion. Define

$$\begin{aligned}\Omega_m(x) = {} & \mathbb{1}[\text{MissionDominant}]\mathbb{1}[\text{FreshProof}]\mathbb{1}[\text{HumanAdmission}] \\ & \cdot \mathbb{1}[\text{AuthorityIssued}]\mathbb{1}[\text{RollbackReady}]\mathbb{1}[\text{ChronicleBound}].\end{aligned} \tag{5}$$

Only when $\Omega_m(x) = 1$ may the release be called Successor Ω, and only for the mission, scope, duration, jurisdiction, data classes and consequence levels actually admitted.

2.3 Mission sovereignty

Mission sovereignty is not legal personhood, self-ownership or unlimited autonomy. It is the institution's ability to preserve mission continuity and bargaining power while substituting models, providers and components. A mission-sovereign architecture owns or controls the mission constitution, job graph, evidence, Chronicle, permissions, portability and economic capture even when some intelligence components are rented.

The sovereignty vector is

$$\mathbf{s}_m = (s_{\text{mission}}, s_{\text{architecture}}, s_{\text{data}}, s_{\text{evidence}}, s_{\text{memory}}, s_{\text{authority}}, s_{\text{portability}}, s_{\text{economics}}). \tag{6}$$

No one coordinate is sufficient. A fully local model can remain non-sovereign if its evidence, verifier, workflow and data rights are controlled elsewhere. A hosted model can participate in a sovereign successor when it is replaceable and the institution retains the mission-level state.

2.4 Authority as a separate state variable

Capability and authority are separate. Let E_t be current evidence and A_t the Authority Envelope. The constitutional constraint is

$$A_t \preceq \phi(E_t), \tag{7}$$

where $\preceq$ orders consequence, scope, duration, reversibility, transaction size and external effect. A candidate may improve in capability while authority remains unchanged. Historical success cannot justify authority after the release, environment, verifier or evidence state materially changes.

Successor Ω law

Specialist ASI is the proven capability state. Successor Ω is the governed institutional state that carries the mission, evidence, memory, rollback and bounded authority.

3 AGI Jobs as Sealed Work-and-Proof Contracts

3.1 Why ordinary tasks are insufficient

A conventional task requests an output. A Bounded AGI Job specifies the complete conditions under which the work may be attempted, evaluated, accepted, reversed and reused. This distinction matters because consequential machine work involves hidden actions, tool calls, external dependencies, data rights, retry costs, human rescue, evidence quality and authority boundaries that are invisible in an ordinary prompt.

Definition 3.1 (Bounded AGI Job). For mission m, job k is the typed contract

$$\mathscr{J}_{m,k} = (o_k, p_k, i_k, \mathscr{D}_k, \mathscr{T}_k, \mathscr{A}_k^+, \mathscr{A}_k^-, b_k, \tau_k, y_k, e_k, v_k, q_k, \rho_k, \chi_k), \tag{8}$$

where o_k is the objective; p_k the accountable principal; i_k acting identity; $\mathscr{D}_k$ authorized data; $\mathscr{T}_k$ tools; $\mathscr{A}_k^+$ permitted actions; $\mathscr{A}_k^-$ prohibited actions; b_k resource budget; τ_k duration or expiry; y_k output schema; e_k evidence obligation; v_k verifier; q_k acceptance predicate; ρ_k rollback route; and χ_k Chronicle eligibility.

The job is sealed when its principal fields are frozen before execution and its resulting proof receipt binds the exact work, inputs, tools, retries, human interventions, outputs, costs and verifier result.

3.2 Job states

A canonical AGI Job state machine is :

Execution is not acceptance. Acceptance is not Chronicle admission. Chronicle eligibility is not authority. Each transition requires its own evidence and accountable actor.

3.3 Job types

A complete successor requires several distinct job families :

Job family	Institutional purpose
Constitution jobs	Freeze mission, beneficiary, incumbent, alternatives, constraints, rights, proof budget and authority ceiling.
Evidence jobs	Acquire, reconcile and provenance-bind mission facts, examples, failures, outcomes and protected cases.
Formation jobs	Build models, policies, world models, tools, retrieval, workflows, specialists, simulators and integrations.
Challenge jobs	Generate anti-theses, adversarial cases, reward attacks, verifier attacks, failure clusters and stop conditions.
Verification jobs	Apply independent scorers, calibrate evidence, count complete burden, test transfer, rollback and degraded modes.
Admission jobs	Produce accountable acceptance, Authority Envelopes, release certificates, Chronicle decisions and revocation conditions.
Renewal jobs	Monitor drift, impair stale releases, re-underwrite the mission, form descendants and run the next fresh proof cycle.

3.4 The proof receipt

Every job emits a proof receipt :

$$\mathscr{R}(\mathscr{J}_{m,k}) = \text{Hash}(\text{job version, inputs, tools, actions, human interventions, cost, evidence, verdict, rollback}). \quad (9)$$

The receipt turns work into attributable institutional evidence. It also exposes hidden retries, external calls and human rescue that could otherwise manufacture false efficiency.

3.5 AGI does not mean one universal worker

An AGI Job may be executed by a frontier model, compact specialist model, deterministic program, agent team, human expert, vendor, laboratory or mixed organization. The name refers to the constitutional unit of machine-era work, not a requirement that one general intelligence perform everything. A job manager or settlement rail may coordinate assignment and payment, but it does not decide mission truth, acceptance, Chronicle admission or authority.

AGI Job law

An AGI Job is bounded mission work with explicit rules, evidence, verification, acceptance criteria and rollback. It prevents "use AI" from becoming an uncontrolled black box.

4 OpenAI Gym Lineage and the Sovereign Mission Environment

4.1 The enduring idea and the current implementation boundary

OpenAI Gym established a standard interface between a learning algorithm and an environment : reset the environment, observe a state-dependent signal, choose an action, receive the next observation and reward, and continue until termination or truncation. The original OpenAI Gym repository is now archived and directs future development to Gymnasium. This paper therefore uses **OpenAI Gym** in the title as the historical and conceptual lineage, while production implementations should use **Gymnasium**-compatible environments and current maintained APIs (OpenAI, 2026 ; Towers et al., 2024).

The distinction matters because the Gym is not an algorithm and does not train a candidate by itself. It is the executable mission constitution that supplies experience, consequence, constraint and evidence. Reinforcement learning, imitation, search, world models, deterministic optimization and human-guided refinement are alternative ways to learn from or operate within that constitution.

Environment law

The Gym manufactures experience ; it does not manufacture truth, certification or authority. Those require independent verifiers, protected fresh work and accountable admission.

4.2 The canonical environment interface

A single-agent environment can be represented as

$$\mathscr{G}_m = \left(\mathscr{X}, \mathscr{O}, \mathscr{A}, \mathscr{T}, \mathsf{r}, \mathsf{c}, \rho, \tau, \mathscr{V}, \mathscr{R}\right), \quad (10)$$

where $\mathscr{X}$ is the hidden mission state, $\mathscr{O}$ the observation structure, $\mathscr{A}$ the permitted action space, $\mathscr{T}$ the mission transition kernel, r a vector-valued mission reward, c non-compensable constitutional constraints, ρ the scenario distribution, τ termination and truncation conditions, $\mathscr{V}$ the independent verifier and $\mathscr{R}$ the proof-receipt function.

The maintained Gymnasium interaction pattern separates termination from truncation :

Listing 1. Conceptual Gymnasium-compatible episode loop.

```
observation, info = env.reset(seed=seed)
while True:
   action = candidate.act(observation)
```

```
    observation, reward, terminated, truncated, info = env.step(action)
    if terminated or truncated:
        break
```

Termination means that the mission reached a genuine terminal state such as accepted success, falsification or critical failure. Truncation means that time, budget, evidence, external availability or authority expired before completion. The distinction is institutionally material because a timed-out candidate is not equivalent to a successful or safely failed candidate.

4.3 From a software benchmark to a corporate mission environment

The sovereign extension adds eight requirements beyond a conventional benchmark :

1. **Mission constitution** : exact objective, beneficiary, incumbent, alternatives and unacceptable failures ;
2. **Role-specific observability** : Producer, Critic, Verifier, Governor, Principal and affected parties may see different information ;
3. **Authority masking** : actions outside the present Authority Envelope are unavailable or technically blocked ;
4. **Vector reward** : quality, value, speed, evidence, reliability, transfer, complete cost and human burden remain visible ;
5. **Hard constraints** : safety, rights, privacy, critical error, budget and governance are not tradable against average reward ;
6. **Adversarial scenarios** : corrupted evidence, distribution shift, provider outage, reward gaming and unauthorized-action traps ;
7. **Proof receipts** : every episode records exact environment, candidate, tools, interventions, costs and verifier verdict ;
8. **Lifecycle partitions** : formation, adversarial, transfer, fresh proof, canary and requalification are separately governed.

4.4 The multiplayer institutional game

The GoalOS Gym is naturally multi-agent. PettingZoo provides standard Agent Environment Cycle and parallel APIs for multi-agent reinforcement-learning environments with role-specific observation and action spaces (Terry et al., 2021). The institutional roles are not symmetrical :

— the **Principal** underwrites, admits and revokes ;
— the **Producer** constitutes proposals and actions ;
— the **Critic** attacks assumptions and searches for failure ;
— the **Verifier** measures outcomes under independent custody ;
— the **Governor** enforces permissions and stop conditions ;
— the **Incumbent** supplies the reference architecture and burden baseline.

Not every role should be trained jointly. The authoritative verifier and admitter must remain outside the candidate's cooperative reward game.

4.5 How the environment is built

The Mission-to-Gym Compiler performs the following sequence :

1. freeze the Mission Constitution and reference alternatives ;
2. enumerate hidden states, observable variables and evidence sources ;
3. define permitted, prohibited and approval-gated actions ;
4. encode deterministic, stochastic, simulated and human-mediated transitions ;
5. define reward vectors and non-compensable hard constraints ;
6. construct normal, edge, adversarial, transfer and tail scenarios ;
7. define verifiers, answer keys, proof receipts and blind custody ;

8. register termination, truncation, refusal, rollback and degraded modes;
9. version, test and sign the EnvironmentSpec;
10. reserve protected cases before formation begins.

4.6 Six Gym families

A production institution separates at least six environment families :

Gym family	Purpose	Candidate visibility
Formation Gym	Teach policies, tools, world models, specialist roles and complete architectures through repeated experience.	Visible and adaptive.
Adversarial Gym	Attack rewards, evidence, tools, verifiers, authority and resilience.	Visible or semi-blind.
Transfer Gym	Test new populations, regimes, properties, providers and failure clusters.	Partially protected.
Fresh-Proof Plane	Evaluate one exact frozen challenger on unseen representative work.	Sealed and non-learning.
Canary Gym	Connect shadow and sandbox results to narrowly reversible external effects.	Limited by authority.
Requalification Gym	Test material changes, drift, expiry, rollback and the successor to the successor.	Version-specific and recurring.

Separation law

The Gym that teaches must not be the Gym that certifies. The candidate may cross the proof firewall; protected information may not return.

5 The Sovereign Mission Gym

5.1 Definition as a constrained partially observable stochastic game

A corporate mission is rarely a clean single-agent Markov decision process. It contains hidden state, partial observations, sequential and simultaneous actions, human and machine roles, external providers, adversaries, budgets, rights, deadlines and irreversible consequences. We therefore define the Mission Gym as a constrained partially observable stochastic game, extending standard environment interfaces (Kaelbling et al., 1998; Towers et al., 2024; Terry et al., 2021; Lanctot et al., 2019).

Definition 5.1 (GoalOS Mission Gym). For mission m, the Mission Gym is

$$\mathcal{G}_m = \left(\mathcal{N}, \mathcal{X}, \{\mathcal{O}_i\}_{i\in\mathcal{N}}, \{\mathcal{A}_i\}_{i\in\mathcal{N}}, \mathcal{T}, \{\mathsf{r}_i\}_{i\in\mathcal{N}}, \mathsf{c}, \rho, \tau, \mathcal{V}, \mathcal{R}\right), \tag{11}$$

where $\mathcal{N}$ is the player set; $\mathcal{X}$ the hidden mission state; $\mathcal{O}_i$ the observation space permitted to player i; $\mathcal{A}_i$ the bounded action space; $\mathcal{T}$ the transition kernel; r_i the mission-dependent payoff vector; c the constitutional constraint vector; ρ the scenario distribution; τ the termination and truncation rules; $\mathcal{V}$ the independent verifier; and $\mathcal{R}$ the proof-receipt function.

The player set may include :

$$\mathcal{N} = \{P, I, B, C, V, G, H, E, R\}, \tag{12}$$

where P is the accountable principal, I the incumbent operator, B the builder or producer, C the critic, V the verifier, G the governor, H affected humans or professional authorities, E external providers or counterparties, and R adversarial reality.

5.2 State, observation and information asymmetry

The true mission state $x_t \in \mathscr{X}$ may include physical, financial, contractual, strategic, organizational and authority conditions that no single actor can fully observe. Player i receives

$$o_t^i \sim O_i(\cdot \mid x_t, \eta_t), \tag{13}$$

where η_t represents sensor noise, missing information, permissions and adversarial distortion. The history available to the player is

$$h_t^i = (o_0^i, a_0^i, \dots, o_t^i), \tag{14}$$

and its policy is

$$\pi_i(a_t^i \mid h_t^i, \theta_i). \tag{15}$$

The Gym must make observation rights explicit. A verifier may see protected ground truth that the producer cannot. The candidate may see the mission input but not scorer internals. The governor may see permissions and budgets but not need access to all confidential business content. The principal may see the final evidence and trade-offs while delegating technical execution.

Design Proposition 5.1 (Information separation). A Mission Gym is constitutionally stronger when the minimum information necessary for each role is sufficient to perform that role, while protected evaluation, release credentials and unnecessary confidential state remain inaccessible.

5.3 Action spaces and the Authority Envelope

The action space is not merely a list of model outputs. It includes requests for evidence, tool calls, internal recommendations, human escalations, sandbox operations and bounded external actions. Let

$$\mathscr{A}_i = \mathscr{A}_i^{\text{analysis}} \cup \mathscr{A}_i^{\text{evidence}} \cup \mathscr{A}_i^{\text{tool}} \cup \mathscr{A}_i^{\text{approval}} \cup \mathscr{A}_i^{\text{external}}. \tag{16}$$

The current Authority Envelope A_t induces an action mask

$$M_{A_t}(a) = \begin{cases} 1, & a \text{ is permitted for the current identity, state and evidence;} \\ 0, & \text{otherwise.} \end{cases} \tag{17}$$

Consequential constraints must be enforced outside the candidate policy. Prompting a candidate not to exceed authority is not an authority system.

5.4 Transition dynamics

The mission evolves according to

$$x_{t+1} \sim \mathscr{T}(\cdot \mid x_t, a_t^{1:n}, w_t), \tag{18}$$

where $a_t^{1:n} = (a_t^1, \dots, a_t^n)$ is the joint action and w_t captures exogenous shocks. In many corporate settings, the transition model is partly deterministic, partly empirical and partly simulated. The Gym may combine :

- historical replay ;
- a deterministic rules engine ;
- a learned world model ;
- a physical or financial simulator ;
- live shadow observations ;
- human-in-the-loop transitions ;
- strategic counterparties or adversarial agents.

The transition model is a major source of Gym basis risk. A high-fidelity interface does not guarantee a high-fidelity world.

5.5 Vector rewards and non-compensable constraints

A single reward scalar invites Goodhart effects and hides mission trade-offs. GoalOS uses a vector reward

$$\mathsf{r}_t = \begin{bmatrix} Q_t & V_t & -T_t & \mathscr{R}_t & E_t & S_t & K_t & -C_t & -H_t \end{bmatrix}^\top, \tag{19}$$

where Q is mission quality, V verified value, T time, $\mathscr{R}$ reliability, E evidence, S sovereignty, K future capacity, C complete cost and H human burden.

Hard constraints remain separate :

$$\mathsf{c}_t = \begin{bmatrix} \text{critical error} \\ \text{unauthorized action} \\ \text{safety violation} \\ \text{rights violation} \\ \text{privacy breach} \\ \text{budget breach} \\ \text{governance regression} \end{bmatrix}. \tag{20}$$

The optimization problem is

$$\max_{\pi} \ \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t U_m(\mathsf{r}_t)\right] \quad \text{subject to} \quad \mathsf{c}_t = 0 \text{ for all hard-gate dimensions.} \tag{21}$$

Many low-severity errors may not compensate for one prohibited catastrophic failure.

5.6 Termination, truncation and refusal

Gymnasium's distinction between termination and truncation is institutionally important (Towers et al., 2024). A mission **terminates** when it genuinely succeeds, fails or reaches a constitutionally defined terminal state. It is **truncated** when time, budget, evidence, authority or external availability expires before completion.

Table 4. Institutional termination and truncation.

Condition	Meaning	Required record
Successful termination	The accepted mission objective was achieved and independently verified.	Proof receipt, cost, evidence, residual uncertainty and authority state.
Failure termination	A critical failure, stop condition or falsification occurred.	Failure class, containment, rollback and Chronicle eligibility.
Time truncation	The deadline arrived before a valid result.	Remaining Proof Debt and whether continuation has positive EVI.
Budget truncation	Proof or operating capital was exhausted.	Complete denominator and new underwriting decision.
Authority truncation	The action required exceeds the current envelope.	Escalation or refusal ; never implicit expansion.
Evidence truncation	Required sources, rights or verifier state expired.	Fail-closed disposition and requalification trigger.

5.7 The proof receipt

Every episode emits an immutable or append-only receipt :

$$\begin{aligned} \mathscr{R}(e) = \text{Hash}\Big(&\text{Gym version, scenario, candidate, observations,} \\ &\text{actions, tools, human interventions}, \mathsf{r}, \mathsf{c}, \text{verdict}\Big). \end{aligned} \tag{22}$$

The receipt must include the exact environment and candidate releases, because neither the Gym nor the successor is stationary.

5.8 Five Gym families

A production-grade institution separates at least five Gym families.

Table 5. The GoalOS Gym stack.

Gym	Purpose	Candidate visibility
Formation Gym	Build tools, policies, world models, specialist organizations and candidate architectures.	Visible ; repeated access permitted.
Adversarial Gym	Attack rewards, tools, evidence, verifier, authority and resilience.	Visible or semi-blind ; designed to expose failure.
Transfer Gym	Test new populations, regimes, properties, industries or failure clusters.	Partially protected ; prevents narrow overfit.
Fresh-Proof Gym	Evaluate one frozen challenger on unseen representative work with protected seeds and scorer internals.	Hidden from candidate and formation roles.
Canary and Requalification Gym	Connect controlled evaluation to shadow, sandbox, reversible bounded action and production monitoring.	Limited according to current authority.

Gym separation law

The Gym that teaches the candidate must not be the Gym that certifies it. Training rewards teach ; protected proof decides ; accountable authority admits.

5.9 Gym maturity

Table 6. Mission Gym maturity model.

Level	State	Evidence
G0	Narrative mission	Objectives and outputs described, but no executable state or verifier.
G1	Replay corpus	Historical episodes, labels and baselines.
G2	Interactive environment	Typed observations, actions, transitions, rewards and termination.
G3	Multi-player institution	Producer, critic, verifier, governor and principal are represented.
G4	Protected proof plane	Blind cases, independent custody, exact releases and confidence gates.
G5	Canary interface	Shadow and reversible external actions with fail-closed controls.
G6	Sovereign Gym asset	Rights-cleared environment, portable APIs, Chronicle integration and requalification.
G7	Recursive succession environment	The Gym and successor co-evolve under protected proof without self-authorization.

6 The AGI Job Contract Calculus

6.1 Why a calculus is required

A job catalogue describes what work exists. A contract calculus specifies when work is admissible, how jobs compose, which evidence is conserved, where failure propagates and why the resulting capability still cannot authorize itself. The relevant comparison is not with ordinary project management. It is with typed interfaces, assume–guarantee contracts, proof-carrying execution and capability-secure systems : each job receives only the information, tools and permissions required by its role, and every accepted transition carries machine-verifiable evidence (Hoare, 1969b ; Lamport, 1994 ; Benveniste et al., 2018 ; Miller et al., 2003).

6.2 Typed signature and operational semantics

For mission m, a job is represented by the typed morphism

$$\mathscr{J}_k : (X_k, E_k, A_k, B_k) \longrightarrow (Y_k, R_k, F_k), \tag{23}$$

where X_k is the authorized input type ; E_k the admissible evidence state ; A_k the action capability ; B_k the resource and time budget ; Y_k the output type ; R_k the proof receipt ; and F_k the declared failure state. A job transition is

$$(x, e, a, b) \overset{\mathscr{J}_k}{\Longrightarrow} (y, r, f) \tag{24}$$

only when the precondition $P_k(x, e, a, b)$ holds. Acceptance requires the guarantee

$$Q_k(y, r) \wedge \mathrm{Verify}_{v_k}(y, r) = 1 \wedge f = \varnothing. \tag{25}$$

The job runtime must fail closed when a required input, right, verifier, budget, credential or source has expired. A plausible output without a valid receipt is an unaccepted output.

Definition 6.1 (Job refinement). Job $\mathscr{J}_j$ refines $\mathscr{J}_i$, written $\mathscr{J}_j \sqsubseteq \mathscr{J}_i$, when it weakens no hard guarantee, expands no prohibited action, preserves the required evidence interface and performs the same mission function with equal or lower complete burden.

Refinement is deliberately stricter than “better benchmark score.” A candidate job that is faster but loses provenance, rollback or independent verification is not a constitutional refinement.

6.3 Assume–guarantee composition

Suppose job i guarantees G_i under assumptions H_i, and job j guarantees G_j under assumptions H_j. Sequential composition is admissible only when

$$G_i \vDash H_j \tag{26}$$

and the output evidence type of i satisfies the input evidence type of j. The composed contract is

$$\mathscr{J}_j \circ \mathscr{J}_i = (H_i,\ G_i \wedge G_j,\ R_i \| R_j), \tag{27}$$

where $R_i \| R_j$ is an append-only receipt chain. This prevents a downstream job from silently assuming facts, rights or authority that no predecessor actually established.

Proposition 6.1 (Evidence conservation). *If every accepted edge in a Job Graph preserves the predecessor receipt and every acceptance predicate is provenance-closed, then every material claim in an accepted terminal output has a directed evidence path to an authorized source or independently produced measurement.*

Démonstration. By induction on a topological order of the accepted Job Graph. Source jobs bind authorized observations to receipts. Each subsequent accepted job preserves predecessor receipts and may add only evidence generated under its own authorized contract. Therefore every terminal material claim traces through accepted edges to an authorized root evidence object. □ □

6.4 The Job Graph as a typed dependency system

Let

$$\mathscr{G}_{J,m} = (V, E, \tau, \pi, \eta, \omega) \tag{28}$$

be the mission Job Graph. Vertices V are jobs ; edges E are typed dependencies ; τ assigns job types ; π assigns accountable principals ; η assigns evidence obligations ; and ω assigns criticality, budget and expiry. The graph may contain parallel branches, joins, challenge loops and bounded repair cycles, but every cycle must have an explicit iteration budget and termination condition.

The runtime exposes four classes of edge :

Edge class	Meaning
Evidence dependency	A downstream job may begin only after a named evidence object has been accepted and remains current.
Control dependency	The successor cannot advance until an accountable principal authorizes the named transition.
Challenge dependency	A critic, red team or verifier must attempt to falsify the predecessor result before acceptance.
Rollback dependency	Failure or impairment activates a known-good predecessor, repair branch or incumbent fallback.

6.5 Completeness and the critical cut

Let $\mathscr{K}_m$ be the set of non-negotiable mission functions. A cut $C \subseteq V$ is constitutionally critical when removal of all jobs in C eliminates evidence for at least one function in $\mathscr{K}_m$. The minimum-weight critical cut is

$$C_m^\star = \operatorname*{arg\,min}_{C:\,\mathrm{Coverage}(\mathscr{G}_{J,m}\setminus C)<1} \sum_{k\in C} \omega_k. \tag{29}$$

It identifies where one missing or invalid job can render the entire candidate incomplete. The board should therefore inspect not only the longest schedule path but the smallest evidence cut capable of invalidating the successor.

Proposition 6.2 (Critical-function blocking). *If a non-negotiable function $r \in \mathscr{K}_m$ has no accepted job path to the frozen candidate manifest, the candidate is constitutionally inadmissible regardless of aggregate performance.*

Démonstration. The Mission Constitution assigns r a hard gate. Absence of an accepted path implies the gate lacks required evidence. Hard gates are non-compensable ; therefore no aggregate reward or benchmark gain can establish admissibility. □ □

6.6 Authority is non-compositional

Capability and evidence can compose. Authority cannot. Let A_k^{form} be the capability granted to job k during formation. Even if every job is accepted,

$$A_m^\Omega \neq \bigvee_{k\in V} A_k^{\mathrm{form}}. \tag{30}$$

Instead,

$$A_m^\Omega \preceq \phi(E_m^{\mathrm{fresh}}, \Gamma_m, H_m), \tag{31}$$

where E_m^{fresh} is current protected proof, Γ_m the constitutional ceiling and H_m the accountable admission record.

Proposition 6.3 (Authority non-composition). *No finite collection of accepted formation jobs entails production authority unless a separate admission transition explicitly maps fresh evidence to an Authority Envelope.*

Démonstration. Formation capabilities are scoped to their job identities, environments, budgets and expiries. The production Authority Envelope is a different state object with a different principal, consequence class and evidence predicate. Since no formation edge has the admitter's release capability, closure under job composition cannot produce that state transition. □ □

6.7 Liveness, bounded recursion and deadlock

A Job Graph is live when every authorized state either reaches an accepted terminal state, reaches a named refusal or rollback state, or reaches a review state within a finite budget. A graph is deadlocked when mutually dependent jobs await evidence or authority that no admissible job can produce. The compiler should detect :

- cycles without a decreasing resource or uncertainty measure ;
- two jobs that each require the other's accepted output ;
- a verifier dependent on a producer-only secret ;

— a rollback target whose rights, version or environment have expired;
— a terminal state with no accountable acceptance owner.

Define the potential

$$\Psi_t = \lambda_U U_t + \lambda_B B_t + \lambda_D D_t, \tag{32}$$

where U_t is unresolved uncertainty, B_t remaining proof budget and D_t unresolved dependency depth. A bounded repair loop is admissible only when $\mathbb{E}[\Psi_{t+1} \mid \Psi_t] < \Psi_t$ or an explicit iteration ceiling forces refusal.

6.8 The constitutional compiler

The Mission Constitution is compiled into a typed Job Graph before execution. The compiler performs five checks :

1. **coverage** : every non-negotiable mission function has at least one admissible job path;
2. **separation** : producer, verifier, admitter and release credentials are not collapsed;
3. **capability safety** : every action is bounded by an external capability mask;
4. **evidence closure** : every acceptance predicate has an independently knowable evidence source;
5. **lifecycle closure** : expiry, refusal, rollback, impairment and requalification are reachable states.

Algorithm 1 Compile a Mission Constitution into a sealed AGI Job Graph

Require : Mission Constitution M, role registry P, capability registry C, verifier registry V
Ensure : Sealed Job Graph $\mathscr{G}_{J,m}$ or a typed refusal

1: derive non-negotiable mission functions $\mathscr{K}_m$ and prohibited effects
2: generate candidate jobs and typed evidence dependencies
3: bind each job to one principal, one identity, one budget and one capability mask
4: insert independent challenge, verification, rollback and expiry jobs
5: reject if coverage < 1 for any hard-gated function
6: reject if producer, verifier or admitter powers are conflated
7: reject if any terminal claim lacks an authorized evidence path
8: reject if a cycle lacks a decreasing potential or finite stop condition
9: freeze schemas, hashes, thresholds, budgets and protected-proof interfaces
10: **return** signed $\mathscr{G}_{J,m}$

Contract-calculus law

AGI Jobs may be composed only when assumptions, guarantees, evidence types, rights and rollback states are compatible. Accepted jobs compose capability and proof; they never compose mission authority.

7 The Typed Job Graph Algebra

7.1 Why a graph, not a checklist

A list of tasks can describe activity while concealing dependency, invalidation, authority and proof. The constitutional manufacturing object is therefore a typed directed graph, borrowing disciplined ideas from axiomatic program specification and event ordering without reducing institutional work to software alone (Hoare, 1969a; Lamport, 1978) :

$$\mathscr{G}_{J,m} = (V_m, E_m, \tau_m, \sigma_m, \pi_m, \nu_m), \tag{33}$$

where V_m contains sealed AGI Jobs; E_m contains evidence, control and dependency edges; τ_m assigns job types; σ_m assigns principals and identities; π_m assigns budgets, permissions and expiry; and ν_m assigns verifier, acceptance and invalidation rules.

Each job has a typed interface

$$\mathscr{J}_k : \; (X_k, P_k, E_k^-; A_k) \longrightarrow (Y_k, E_k^+, F_k), \tag{34}$$

where X_k is the authorized input state; P_k the precondition; E_k^- predecessor evidence; A_k the permitted action set; Y_k the deliverable; E_k^+ new evidence; and F_k the declared failure or refusal state. A job is executable only when its preconditions, rights, evidence freshness and authority are satisfied.

7.2 Three kinds of edges

The graph distinguishes three edge classes :

Edge	Meaning
Evidence edge $i \to_E j$	Job j may rely on a named, versioned and current evidence object emitted by job i.
Control edge $i \to_C j$	Job i authorizes, blocks, narrows, expires or rolls back job j without supplying substantive mission evidence.
Invalidation edge $i \dashv j$	Failure, expiry, rights loss or verifier impairment at job i invalidates all named claims, permissions or descendants at job j.

Conflating these edges is dangerous. Evidence does not itself authorize execution; authorization does not establish truth; and a valid output does not automatically enter Chronicle.

7.3 The triadic completeness condition

Let $\mathscr{F}_m$ be the set of required formation, challenge, verification, admission and renewal job families for mission m. Define formation coverage

$$\mathrm{Cov}_{\mathrm{form}}(\mathscr{G}_{J,m}) = \frac{\sum_{f\in\mathscr{F}_m} w_f\, \mathbb{1}[\exists k : \tau(k) = f \wedge q_k = 1]}{\sum_{f\in\mathscr{F}_m} w_f}. \tag{35}$$

Coverage of one means that the declared manufacturing obligations have been satisfied. It does *not* establish mission dominance. It establishes only that the candidate is constitutionally complete enough to be submitted to fresh proof.

Proposition 7.1 (Formation completeness is not proof sufficiency). *If* $\mathrm{Cov}_{\mathrm{form}}(\mathscr{G}_{J,m}) = 1$, *then the graph may produce a frozen candidate manifest. It does not imply* $\alpha_m > 0$, *Specialist ASI, Chronicle admission or any Authority Envelope.*

Démonstration. Formation coverage is a statement about the presence and acceptance of manufacturing obligations. Mission dominance additionally depends on protected comparative outcomes against the incumbent and best credible alternative, complete-cost accounting and hard gates. Accountable admission additionally depends on rights, authority, monitoring, expiry, rollback and human decision. These states are logically distinct. □

7.4 Authority is non-compositional

Let A_k be the bounded permissions granted to job k. A naive system might infer successor authority from $\bigcup_k A_k$. GoalOS prohibits that inference. The rule is compatible with the broader security principle that permitted information or actions must follow an explicit lattice rather than emerge accidentally from composition (Denning, 1976).

Proposition 7.2 (No authority by union). *For any job graph* $\mathscr{G}_{J,m}$, *the production Authority Envelope* A_m^Ω *is not the union, sum or transitive closure of job permissions :*

$$A_m^\Omega \neq \bigcup_{k\in V_m} A_k. \tag{36}$$

Instead,

$$A_m^\Omega = \mathsf{IssueAuthority}_m(E_t, R_m, P_m), \tag{37}$$

where E_t *is current proof,* R_m *the mission risk and rollback state, and* P_m *the accountable admission decision.*

Démonstration. Job permissions exist to execute bounded formation and verification work. Their union can contain mutually incompatible identities, data classes, tools and consequence levels. A separate Authority Envelope is required to establish one coherent, scoped and revocable production state. Otherwise privilege can escalate through decomposition alone. □

No privilege escalation by composition

AGI Jobs compose capability and evidence. They do not compose authority. No sequence, parallel set or recursive family of jobs may manufacture a larger Authority Envelope than a separately accountable principal issues from current proof.

7.5 Evidence invalidation and Proof Debt propagation

Let $\mathrm{Dep}(j)$ be the evidence ancestors of job j. If ancestor i expires or is falsified, downstream claims depending on i inherit Proof Debt :

$$D_P(j,t+1) = D_P(j,t) + \sum_{i\in\mathrm{Dep}(j)} \omega_{ij}\,\mathbb{1}[\mathrm{Invalid}(i,t)], \tag{38}$$

where ω_{ij} is the declared dependence weight. A high-dependence job graph therefore needs automatic impairment, not merely a periodic narrative review.

7.6 Critical cuts and minimum viable proof

A cut set $K \subseteq V_m$ is critical when removing or invalidating K disconnects the candidate from a required constitutional state such as independent verification, rollback, rights clarity or human admission. Define the mission-critical cut family

$$\mathscr{K}_m = \{K : \mathscr{G}_{J,m} \setminus K \text{ cannot reach FreshProof or AccountableAdmission}\}. \tag{39}$$

The smallest such cuts identify control points whose compromise can invalidate the whole successor. Typical one-node cuts include the authoritative verifier, rights constitution, rollback package or accountable admitter. The institution should reduce single-point dependence where feasible, while preserving separation of powers.

7.7 Parallelism and serial constitutional gates

Formation jobs may execute in parallel when their data rights, budgets and evidence states are independent. Constitutional gates remain serial :

$$\text{Underwrite} \prec \text{Execute} \prec \text{Accept} \prec \text{Admit} \prec \text{Allocate}. \tag{40}$$

Parallel compute cannot collapse those decisions. The fastest graph is not the graph with the fewest human decisions; it is the graph that removes avoidable latency while retaining the decisions that preserve mission truth and accountable authority.

8 The Constitutional Job Compiler

8.1 From mission prose to executable institutional work

A mission statement is not yet an executable successor programme. The Constitutional Job Compiler converts a frozen Mission Constitution into a typed, dependency-aware portfolio of AGI Jobs, proof obligations, information boundaries and stop conditions. Its output is not merely a task list. It is a machine-readable manufacturing constitution whose structure can be inspected before capital, data or authority are exposed.

Let the frozen mission constitution be

$$\mathfrak{C}_m = (o_m, b_m, I_m, \mathscr{B}_m, \Theta_m, \mathscr{R}_m, \mathscr{H}_m, \bar{A}_m, \tau_m), \tag{41}$$

where o_m is the objective; b_m the beneficiary; I_m the incumbent; $\mathscr{B}_m$ the best-credible-alternative set; Θ_m the success and failure thresholds; $\mathscr{R}_m$ the rights and data boundary; $\mathscr{H}_m$ the non-negotiable hard constraints; $\bar{A}_m$ the maximum constitutional authority ceiling; and τ_m the expiry and review schedule.

The compiler is the map

$$\mathsf{Compile}_J : \mathfrak{C}_m \longmapsto (\mathscr{G}_{J,m}, \mathscr{I}_m, \mathscr{P}_m, \mathscr{S}_m), \tag{42}$$

where $\mathscr{G}_{J,m}$ is the typed Job Graph; $\mathscr{I}_m$ the role-specific information-flow policy; $\mathscr{P}_m$ the proof-obligation registry; and $\mathscr{S}_m$ the stop, rollback and invalidation policy.

Compiler law

The compiler may expand a mission into bounded work. It may not expand the mission, relax a hard gate, create authority or select its own acceptance threshold.

8.2 A job type system

Each AGI Job carries three distinct types :

$$\mathrm{Type}(\mathscr{J}_k) = \left(\tau_k^{\mathrm{work}}, \tau_k^{\mathrm{evidence}}, \tau_k^{\mathrm{authority}}\right). \tag{43}$$

The work type identifies what transformation is attempted; the evidence type identifies what proof object must be returned; the authority type identifies the maximum class of external effect permitted during that job. These types must not be conflated.

Type family	Examples and constitutional meaning
Work type	Extract, reconcile, classify, plan, simulate, generate, challenge, verify, admit, monitor, rollback or revoke.
Evidence type	Source-linked record, executable test, signed measurement, calibrated score, human decision, cryptographic commitment or proof receipt.
Authority type	Read only, recommend, sealed sandbox, reversible bounded action or consequential action under dual control.
Criticality type	Advisory, material, safety-critical, rights-affecting, capital-affecting or release-critical.
Secrecy type	Public, internal, confidential, protected proof, privileged, regulated or professionally restricted.
Lifecycle type	Formation, fresh proof, canary, operation, Chronicle admission, requalification, impairment or retirement.

A well-typed edge $\mathscr{J}_i \to \mathscr{J}_j$ requires that the output evidence type of i satisfies the input obligation of j, the rights scope remains valid, the evidence has not expired, and the information policy permits the transfer. A syntactically convenient edge can still be constitutionally invalid.

8.3 Information-flow control

Let $\ell(x)$ denote the information label of object x and let $\preceq_I$ be the permitted information-flow relation. A job may consume evidence e_i only when

$$\ell(e_i) \preceq_I \ell(\mathscr{J}_j) \quad \text{and} \quad \mathrm{RightsValid}(e_i, j) = 1. \tag{44}$$

This supports role-specific views : the Producer can receive formation evidence while remaining unable to inspect protected fresh-proof cases; the Verifier can inspect sealed cases while lacking release credentials; the Admitter can receive a signed verdict without receiving the candidate's private training state.

The proof firewall therefore becomes a non-interference condition. If Y_F denotes formation-visible outputs and Z_P protected proof information, then

$$I(Y_F; Z_P \mid \mathfrak{C}_m) \leqslant \varepsilon_{\mathrm{leak}}, \tag{45}$$

for a preregistered tolerance $\varepsilon_{\mathrm{leak}}$, with any material leakage invalidating the affected release. The expression is an operational criterion, not a claim that mutual information can always be estimated perfectly in production.

8.4 Proof-carrying work

A job result is not merely an output y_k. It is a proof-carrying work object

$$\mathsf{PCW}_k = (y_k, \pi_k, \nu_k, \delta_k), \tag{46}$$

where π_k binds inputs, tools, actions, retries and human interventions; ν_k is the verifier verdict and residual uncertainty; and δ_k identifies dependencies whose later impairment invalidates the result.

The acceptance predicate is

$$q_k(\mathsf{PCW}_k) = \mathbb{1}[\text{schema}]\mathbb{1}[\text{evidence}]\mathbb{1}[\text{rights}]\mathbb{1}[\text{hard gates}]\mathbb{1}[\text{verifier}]\mathbb{1}[\text{rollback ready}]. \tag{47}$$

A persuasive answer with no attributable evidence has $q_k = 0$ even when its substantive conclusion later proves correct. GoalOS distinguishes lucky correctness from institutionally admissible capability.

8.5 Authority is non-compositional

Let A_k^{form} be the maximum action class granted to job k during formation. The production Authority Envelope of the successor is not the union, join or transitive closure of those permissions.

Proposition 8.1 (Authority non-composition). *For a Job Graph $\mathscr{G}_{J,m}$,*

$$A^{\Omega}_{m,t} \not\equiv \bigvee_{k \in V_J} A_k^{\text{form}}. \tag{48}$$

Instead,

$$A^{\Omega}_{m,t} \preceq \phi\left(E^{\text{fresh}}_{m,t}, \bar{A}_m, \mathsf{AdmissionRecord}_{m,t}\right). \tag{49}$$

Démonstration. Formation permissions are granted for different purposes, identities, environments, durations and data classes. Their aggregation can create an action class that was never evaluated as a complete operational capability. Fresh proof evaluates one frozen release under one declared action surface, and accountable admission issues a new envelope for that exact release. Therefore formation permissions cannot imply production authority without violating the separation between execution, acceptance and admission. □

8.6 Invalidation propagation

Let $\Delta^- e_i$ denote the impairment or expiry of predecessor evidence. Define the dependent closure

$$\mathrm{Desc}(i) = \{j : \text{a directed path exists from } i \text{ to } j\}. \tag{50}$$

The invalidation operator is

$$\mathsf{Invalidate}(i) = \{i\} \cup \{j \in \mathrm{Desc}(i) : \mathrm{DependencyMaterial}(i,j) = 1\}. \tag{51}$$

A downstream result may remain usable only when an independent proof object severs the affected dependency. Otherwise the system marks the result impaired, narrows authority or triggers reproof.

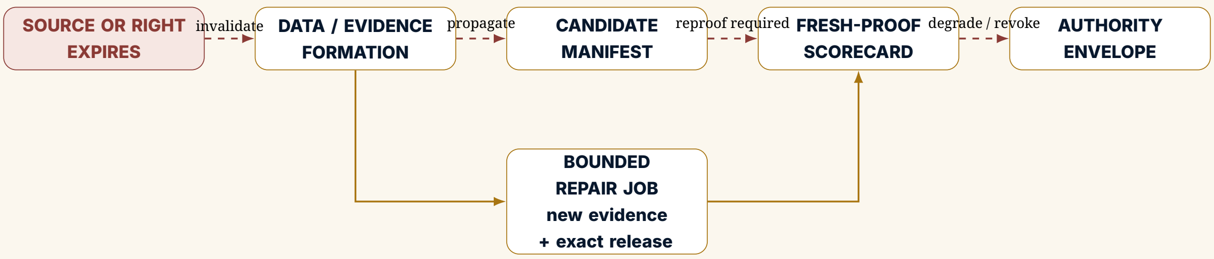

FIGURE 2. Illustrative invalidation propagation. Evidence expiry travels through material dependencies; it does not silently preserve proof, Chronicle status or authority.

8.7 Minimal assurance cut sets

For a mission-critical failure f, an assurance cut set is a set of jobs whose failure can disconnect the graph from an admissible successor state. Let $\mathscr{C}_f$ be the family of such cut sets. A design is fragile when one low-observability job appears in many minimal cuts.

Define assurance concentration

$$\mathrm{AC}(\mathscr{G}_{J,m}) = \max_{k \in V_J} \frac{|\{C \in \mathscr{C} : k \in C\}|}{|\mathscr{C}|}. \tag{52}$$

High assurance concentration calls for independent redundancy, diverse verification or a narrower mission. The objective is not redundant bureaucracy; it is to prevent one opaque component from becoming the unseen constitutional root of the successor.

8.8 Compiler algorithm

Algorithm 2 Constitutional Job Compilation

Require : Frozen mission constitution $\mathfrak{C}_m$, incumbent I_m, alternative set $\mathscr{B}_m$
Ensure : Typed Job Graph $\mathscr{G}_{J,m}$ or a refusal record

```
Verify objective, beneficiary, rights, hard constraints, proof budget and authority ceiling
if any constitutional field is unresolved then
    return REFUSE with Proof Debt register
end if
Generate materially distinct formation and evidence obligations
Assign work, evidence, authority, secrecy, criticality and lifecycle types
Insert independent challenge, verification, rollback and admission jobs
Build dependency edges and role-specific information-flow policy
Compute coverage, assurance cut sets, critical path and expected value of information
Reject privilege-escalation paths and protected-proof leakage
Emit sealed job manifests, verifier obligations and stop-loss schedule
return (G_{J,m}, I_m, P_m, S_m)
```

Compiler outcome

The output is not "an AI workflow." It is a typed constitutional manufacturing plan whose work, evidence, information, capital and authority boundaries can be inspected before the successor exists.

9 The Triadic Compiler and Proof-Carrying Execution

9.1 From a mission to a governed executable institution

The triadic compiler transforms one constituted mission into three synchronized graphs :

1. a **Job Graph** describing work, dependencies, evidence and invalidation;
2. an **Environment Graph** describing state, observations, actions, transitions, rewards, constraints and scenario families; and
3. an **Authority Graph** describing principals, permissions, approvals, blocked actions, expiry, rollback and revocation.

The graphs are linked but not merged. Evidence may flow from Jobs and the Gym into proof. Authority may flow only from accountable admission into operation.

9.2 Canonical machine-readable objects

Object	Purpose
`MissionConstitution`	Freezes objective, beneficiary, incumbent, alternatives, hard gates, rights, proof capital and authority ceiling.
`AGIJobContract`	Specifies bounded work, identity, inputs, tools, prohibitions, budget, evidence, verifier, acceptance, expiry and rollback.

`JobGraphManifest`	Records evidence, control and invalidation edges among jobs.
`EnvironmentSpec`	Versions hidden state, observations, actions, transitions, rewards, constraints, scenario distributions and episode endings.
`CandidateSuccessor Manifest`	Identifies the complete exact release : models, policies, world models, tools, workflow, humans, verifier, cost and dependencies.
`RunProof`	Carries episode provenance, actions, tools, interventions, rewards, violations, cost and verifier verdict.
`FreshSuccessor Scorecard`	Compares one frozen candidate with the incumbent and best alternative on protected work.
`AuthorityEnvelope`	Defines permitted objectives, data, tools, actions, budgets, approvals, stop conditions, expiry and rollback.
`ChronicleRecord`	Admits only evidenced, attributable, rights-cleared, versioned, bounded and challengeable capability.
`Requalification Record`	Records material changes, impairment, repair, rollback, revocation and next proof date.

9.3 Proof-carrying work and action

Each completed job emits a proof receipt

$$\mathscr{R}(j) = \mathrm{Hash}(m, j, \mathrm{inputs}, \mathrm{tools}, \mathrm{actions}, \mathrm{interventions}, \mathrm{cost}, \mathrm{evidence}, \mathrm{verdict}, \mathrm{version}). \tag{53}$$

A consequential action is admissible only when its receipt, current evidence and Authority Envelope agree. This converts proof from a retrospective report into a precondition of action.

9.4 Non-composition of authority

Let A_j be the bounded permissions required by job j. The successor authority is not their union :

$$A_m^{\Omega} \neq \bigcup_{j \in \mathscr{G}_{J,m}} A_j. \tag{54}$$

Job permissions exist to manufacture and test candidacy. Successor authority is separately reconstituted from fresh evidence, mission risk, rights, rollback readiness and accountable admission.

9.5 Invalidation and reproof

If a source expires, a verifier fails, a provider changes, a model or tool is replaced, or a material distribution shift occurs, invalidation propagates through the Job Graph to affected claims, proof receipts and authority classes. The institution must be able to identify :

— which jobs and evidence objects are invalidated ;
— which Specialist ASI claims no longer have full Proof Coverage ;
— which actions must degrade, pause or revoke ;
— what smallest repair job and fresh proof restore admissibility.

Triadic compiler law

Jobs may compose capability. The Gym may compose experience. Proof may compose evidence. Authority never composes automatically ; it is separately granted, continuously collateralized and revocable.

10 The Job Graph and Compositional Successor Manufacture

10.1 From a list of tasks to a constitutional dependency graph

The complete job portfolio is not a flat checklist. It is a directed, typed dependency graph

$$\mathcal{G}_{J,m} = (V_J, E_J, \lambda, \kappa), \tag{55}$$

where vertices are AGI Jobs, edges encode evidence and dependency requirements, λ assigns job types and principals, and κ assigns criticality, budget, authority and proof obligations.

A job may begin only when its predecessor evidence is valid, rights-cleared and current. A downstream acceptance may be invalidated when a predecessor is impaired. The graph therefore functions as both manufacturing plan and control surface.

10.2 A canonical manufacturing graph

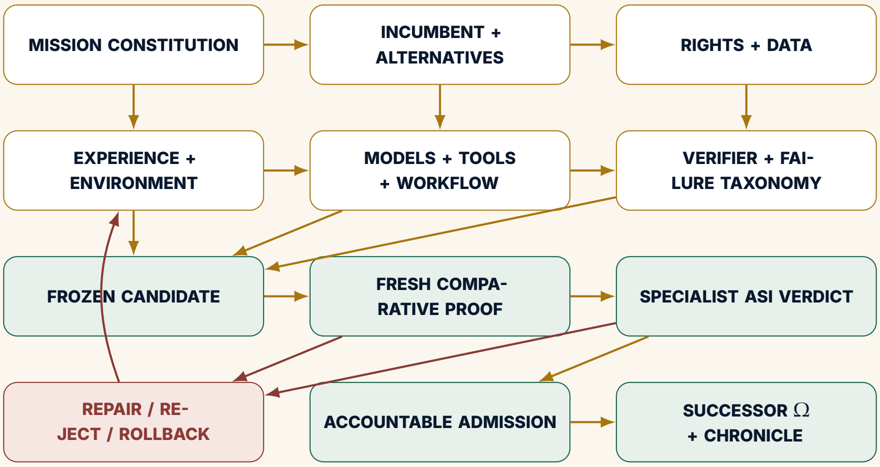

FIGURE 3. The Successor–Job manufacturing graph. Formation creates candidacy; protected proof creates a verdict; accountable admission creates the Successor Ω state.

10.3 Completeness

Let $\mathcal{K}_m$ be the set of mission-critical institutional functions : model, data, orchestration, tools, verifier, workflow, rights, integration, evidence, Chronicle, rollback and authority. Define coverage

$$\mathrm{Coverage}(\mathcal{G}_{J,m}) = \frac{\sum_{r\in\mathcal{K}_m} w_r \mathbb{1}[\exists\, \mathcal{J}_k \text{ accepted for } r]}{\sum_{r\in\mathcal{K}_m} w_r}. \tag{56}$$

Candidate completeness requires full coverage for every function marked non-negotiable by the Mission Constitution. High model performance cannot compensate for an absent verifier, missing rights, untested rollback or undefined human authority.

10.4 Compositionality without authority leakage

Job permissions do not automatically aggregate into successor authority. Let A_k be the action class permitted to job k during formation. The authority of the complete successor is not

$$A^{\Omega} = \bigvee_k A_k.$$

Instead,

$$A^{\Omega}_{m,t} \preceq \phi\left(E^{\mathrm{fresh}}_{m,t}, \text{mission ceiling, admission record}\right). \tag{57}$$

A job may use a tool inside a sealed sandbox without making that tool available in production. A verifier may inspect protected data without disclosing it to the producer. A formation worker may generate a descendant without holding release credentials. This non-compositional authority rule prevents the job graph from becoming a covert privilege escalation path.

10.5 Critical path and proof capital

The job graph creates a critical path through evidence and formation. Proof capital should be allocated to the job or dependency whose resolution most changes the succession decision. For diagnostic job k, define

$$\mathrm{EVI}_k = \mathbb{E}\left[\max_{a\in\mathscr{A}} U(a \mid e_k)\right] - \max_{a\in\mathscr{A}} \mathbb{E}[U(a)] - C(\mathscr{J}_k). \tag{58}$$

The next job should maximize positive expected value of information subject to rights, safety, time and stop-loss constraints. This makes the job graph a real-options instrument rather than an unlimited research queue.

Compositionality law

Accepted AGI Jobs may compose capability and evidence. They do not compose authority. The complete successor receives authority only through a separate admission record tied to current fresh proof.

11 The Complete Successor Is Larger Than the Policy

11.1 The policy fallacy

The dominant technical abstraction in sequential decision-making is a policy π that maps observations or histories to actions. This abstraction is necessary but institutionally incomplete. A policy does not, by itself, establish :

- whether the mission objective is legitimate and frozen ;
- whether the observations are rights-cleared and sufficiently complete ;
- whether tools and external systems are permitted ;
- whether evidence is preserved ;
- whether the output is independently verified ;
- whether humans were required to rescue the result ;
- whether a recommendation may become an action ;
- whether the system can roll back or degrade safely ;
- whether learning may influence future missions ;
- whether a later release remains the same institutional object.

The complete successor must therefore be represented as an institution rather than a policy.

Definition 11.1 (Complete Successor Institution). For mission m, a complete successor is

$$\gg_m = (M, D, W, H, T, R, V, P, O, E, C, A, B)\,, \tag{59}$$

where M is the model portfolio ; D rights-cleared data and examples ; W the world model and knowledge representation ; H the harness, orchestration and human roles ; T tools, retrieval, code and simulators ; R rules, rights, security and resilience ; V the verifier system ; P the operating policy portfolio ; O workflow and organizational integration ; E the evidence state ; C Chronicle ; A the Authority Envelope ; and B rollback, fallback and lifecycle controls.

A single policy is a component :

$$\pi \in P \subset \gg_m\,. \tag{60}$$

11.2 The institutional stack

11.3 World model and mission model

The successor's internal world model $\mathscr{W}_t$ is not identical to the Gym's authoritative hidden state x_t. The successor estimates :

$$q_\theta(x_t \mid h_t) = \Pr(x_t \mid h_t; \theta), \tag{61}$$

while the Gym or real environment governs the actual transition. A strong successor maintains explicit separation among :

Figure 4. The complete Successor is larger than the acting policy.

— observed fact ;
— externally asserted claim ;
— derived calculation ;
— model inference ;
— counterfactual scenario ;
— unresolved uncertainty.

This separation is essential in acquisition underwriting, building operations, financial reconciliation and other missions where persuasive synthesis can conceal missing evidence.

11.4 Policy portfolio rather than monolith

A complete successor may route among several policies :

$$\pi^\star(a \mid h) = \sum_{k=1}^{K} g_k(h)\pi_k(a \mid h), \tag{62}$$

where $g_k(h)$ is a governed routing distribution. The portfolio may include :

— a general frontier model for broad reasoning ;
— a local specialist model for confidential classification ;
— deterministic code for calculations and invariants ;
— search or planning for sequential choices ;
— a critic with different data or model lineage ;
— a human professional for reserved judgment.

The routing policy is itself subject to proof. A successor can fail through incorrect routing even when every individual specialist is competent.

11.5 Verifier independence

The verifier is not merely another prompt asking whether the answer looks correct. It may include :

— deterministic invariants ;
— hidden ground truth ;
— source and provenance checks ;
— adversarial alternative generation ;
— execution tests ;

— independent models or methods ;
— human or professional adjudication ;
— post-outcome feedback.

Let Y be the true acceptance state and $\hat{Y}_V$ the verifier decision. The verifier must be measured for both false acceptance and false rejection :

$$\mathrm{FAR}_V = \Pr(\hat{Y}_V = 1 \mid Y = 0), \qquad \mathrm{FRR}_V = \Pr(\hat{Y}_V = 0 \mid Y = 1). \tag{63}$$

A verifier that is too permissive launders unsupported capability into authority. A verifier that is too restrictive destroys option value and forces repeated work. Chronicle precision depends on both.

11.6 Human roles are components of the architecture

The human is not an unspecified safety fallback. The complete architecture must state which human roles remain responsible for :

— mission purpose ;
— professional judgment ;
— exception resolution ;
— stakeholder communication ;
— legal or fiduciary authority ;
— acceptance and admission ;
— incident response and revocation.

Human burden is a denominator, not a hidden subsidy. Let $H_m(S)$ include review time, escalation, rescue, training, contestation and professional exposure. A candidate that appears autonomous only because humans repair its failures has not established lower burden.

11.7 Lifecycle and degraded modes

The complete successor defines at least six modes :

$$\text{Full} \longrightarrow \text{Restricted} \longrightarrow \text{Recommend only} \longrightarrow \text{Read only} \longrightarrow \text{Incumbent fallback} \longrightarrow \text{Emergency stop}. \tag{64}$$

The transition policy among modes is part of the successor, not an afterthought. It must respond to evidence expiry, provider failure, distribution shift, security incident, budget exhaustion, critical error and authority loss.

Complete successor law

A policy can act in the Gym. Only a complete Successor Institution can inherit a mission. Models produce capability ; the institution produces evidence, continuity, accountability and bounded authority.

12 Compositional Assurance and Proof-Preserving Manufacture

12.1 Why local job success is not enough

A successor may contain many individually successful jobs while remaining globally inadmissible. Local accuracy can conceal missing coverage, correlated verifier error, stale dependencies, contradictory outputs, authority leakage or an unsafe interaction among components. Compositional assurance asks whether accepted job results form a coherent, current and independently challengeable complete institution.

Let $Q_k \in \{0, 1\}$ indicate acceptance of job k, let D_{ij} indicate a material dependency, and let G_m denote the mission's non-negotiable constitutional gate. A naive completion rule

$$\prod_{k \in V_J} Q_k = 1 \tag{65}$$

is insufficient because it ignores dependency validity, role separation and complete-system interaction.

12.2 Proof-preserving composition

Define the composition predicate

$$\begin{aligned}\mathsf{ComposeOK}(\mathscr{G}_{J,m}) = {} & \mathsf{Coverage} \wedge \mathsf{DependencyValidity} \wedge \mathsf{Rights} \wedge \mathsf{VerifierIndependence} \\ & \wedge \mathsf{InformationSeparation} \wedge \mathsf{Rollback} \wedge \mathsf{NoAuthorityLeakage}.\end{aligned} \tag{66}$$

Proposition 12.1 (Proof-preserving composition). *Suppose every accepted job emits a valid proof-carrying work object, every material edge is current and rights-valid, independent verification covers the mission-critical cut sets, and* $\mathsf{ComposeOK}(\mathscr{G}_{J,m}) = 1$. *Then the assembled candidate is eligible to be frozen for fresh comparative proof. It is not yet Specialist ASI and holds no inherited production authority.*

Démonstration. The premises establish a complete, attributable and internally coherent candidate release. They do not establish comparative mission dominance because formation evidence may be reused, selected or overfit. They also do not establish accountable admission. Eligibility for freezing is therefore the strongest valid conclusion. □

12.3 No certification by graph closure

Even a fully accepted Job Graph cannot certify itself. Let $\mathsf{Closure}(\mathscr{G}_{J,m})$ denote completion of all declared internal obligations. Then

$$\mathsf{Closure}(\mathscr{G}_{J,m}) \nRightarrow \mathsf{SpecialistASI} \quad \text{and} \quad \mathsf{Closure}(\mathscr{G}_{J,m}) \nRightarrow \mathsf{SuccessorOmega}. \tag{67}$$

Fresh proof and accountable admission are external state transitions.

12.4 Verifier correlation and evidence diversity

Multiple verifiers do not create independence when they share the same model family, data, prompt, scoring heuristic or organizational incentive. Let $Z_1, \dots, Z_r$ be verifier error indicators with correlation matrix Σ_V. An effective verifier count can be approximated by

$$r_{\text{eff}} = \frac{r^2}{\mathbf{1}^\top \Sigma_V \mathbf{1}}. \tag{68}$$

When all verifier errors are perfectly correlated, $r_{\text{eff}} = 1$ regardless of nominal committee size. The assurance plan should diversify methods, evidence sources, incentives and failure modes—not merely model endpoints.

12.5 Conservative job-graph value

Let V_k denote the accepted marginal contribution of job k, C_k its complete burden, R_k residual risk and Γ_{ij} interaction value or interaction loss. A conservative Job-Graph contribution is

$$\widehat{V}_J = \sum_k (V_k - C_k - R_k) + \sum_{(i,j)\in E_J} \Gamma_{ij} - D_J - \varepsilon_J, \tag{69}$$

where D_J is unresolved graph-level Proof Debt and ε_J is composition uncertainty. The term Γ_{ij} can be negative : a fast Producer paired with a weak Verifier may reduce complete value even when each component scores well alone.

12.6 Job-level Shapley attribution

For post hoc diagnosis rather than authority allocation, the marginal contribution of job k to accepted mission value can be estimated using a Shapley-style attribution :

$$\varphi_k = \sum_{S \subseteq V_J \setminus \{k\}} \frac{|S|!(K - |S| - 1)!}{K!} \big[v(S \cup \{k\}) - v(S) \big]. \tag{70}$$

This can identify whether value comes from better evidence, stronger challenge, lower burden or improved rollback. It must not be used to infer that a high-value component deserves greater authority.

12.7 Monotone invalidation

Proposition 12.2 (Authority cannot increase under evidence loss). *If evidence state $E_{t+1} \preceq_E E_t$, then the admissible authority satisfies*

$$\phi(E_{t+1}) \preceq_A \phi(E_t). \tag{71}$$

Démonstration. The Authority Envelope is collateralized by current evidence. Evidence expiry, impairment or loss cannot justify a larger consequence, scope, duration or permission set. At best authority is unchanged when the impaired evidence was immaterial ; otherwise it narrows or fails closed. □

12.8 Bounded recursive manufacture

Let generation g contain Job Graph $\mathscr{G}^{(g)}$, candidate $S^{(g)}$, proof state $E^{(g)}$ and authority $A^{(g)}$. The successor operator proposes descendants :

$$\{(\mathscr{G}_i^{(g+1)}, S_i^{(g+1)})\}_{i=1}^{n} = \mathfrak{R}(\mathscr{G}^{(g)}, S^{(g)}, Chronicle_g). \tag{72}$$

No descendant inherits $E^{(g)}$ or $A^{(g)}$. It may reuse rights-cleared environments, accepted tools, failure maps and job templates, but every material claim and action class is re-frozen and re-proven.

Assurance law

Job completion manufactures a candidate. Proof-preserving composition makes the candidate eligible for freezing. Only fresh comparative proof can establish Specialist ASI, and only accountable admission can create Successor Ω.

PART II

Formal Assurance, Evidence and the Executable Institution

A successor is promoted by a proof procedure, not by a score, narrative or model reputation

The ceiling edition converts constitutional principles into conditional formal guarantees, promotion statistics, basis-risk reserves and explicit invalidation conditions.

SUCCESSOR MANIFOLD → GRADIENT → SEIZE → BOUNDED JOBS → SPECIALIST ASI → SUCCESSOR Ω

13 Formal Assurance : Axioms, Promotion Statistics and Robust Admission

13.1 Purpose and epistemic boundary

This section strengthens the design from a collection of rules into a conditional assurance system. The results below are not universal laws of organizations, intelligence or markets. They are theorems and propositions *inside a declared GoalOS constitution* : if the mission, candidate set, evidence relation, verifier, hard gates and authority map satisfy the stated assumptions, then the corresponding institutional conclusion follows. If the assumptions fail, the guarantee fails with them.

Formal-assurance rule

No theorem can rescue a misconstituted mission, a captured verifier, leaked protected cases, an omitted tail risk or a falsely declared authority boundary. Formal assurance begins by making those assumptions explicit, testable and revocable.

13.2 Eight constitutional axioms

Let m denote a constituted mission ; x an exact complete candidate release ; $\mathscr{B}_m$ the best credible alternative set ; E_t the current evidence state ; A_t the active Authority Envelope ; and $\mathscr{C}_m(x)$ the conjunction of non-negotiable mission gates.

Axiom	Constitutional statement
A1 Mission identity	The objective, beneficiary, incumbent, alternative set, evaluation distribution, critical failures and authority ceiling are frozen before protected evaluation.
A2 Release identity	A proof applies only to the exact model, data, tool, workflow, verifier, policy, runtime, provider and authority configuration recorded in the Candidate Successor Manifest.
A3 Independent custody	Protected cases, hidden labels, scorer internals and release credentials remain outside candidate formation and outside the candidate's own control.
A4 Hard-gate non-compensation	No amount of average reward, speed, revenue or low-severity success compensates for a prohibited catastrophic failure or a failed constitutional gate.
A5 Complete denominator	Mission value counts construction, operation, proof, human rescue, integration, security, rights, dependency, requalification, rollback and unwind.
A6 Authority separability	Capability, acceptance, Chronicle admission and authority are distinct state variables controlled by separately attributable decisions.
A7 Freshness	Material change to the mission, release, environment, verifier or Authority Envelope expires the affected proof claims.
A8 Revocability	Every admitted successor has a known-good fallback, expiry, impairment condition and revocation route.

13.3 Exact release identity and proof validity

Define the release identity as the cryptographic commitment

$$\mathsf{RID}(x) = \mathrm{Hash}(m, M, D, H, T, V, W, R, I, E, C, A, \mathrm{runtime}, \mathrm{provider}). \tag{73}$$

A proof record P is valid for release x only when

$$\mathsf{ValidProof}(P, x) = \mathbb{1}[\mathsf{RID}(P) = \mathsf{RID}(x)]\,\mathbb{1}[\mathsf{Custody}(P)]\mathbb{1}[\mathsf{Fresh}(P)]\mathbb{1}[\mathsf{Reproducible}(P)]. \tag{74}$$

This prevents a later release from borrowing the identity of an earlier release while changing a component that can alter the mission result.

Proposition 13.1 (No proof by family resemblance). *If* $\mathsf{RID}(x') \neq \mathsf{RID}(x)$ *for a mission-relevant component, a proof of* x *does not establish the corresponding claim for* x' *unless a declared invariance argument is independently validated.*

Démonstration. A mission claim is conditional on the release that generated the observed outcomes. A mission-relevant difference can change the transition distribution, evidence quality, action set, failure modes or authority exposure. Without a validated invariance argument, the likelihood of the observed evidence under x' is not identified by the evidence generated under x. The inherited claim is therefore unsupported. □

13.4 Existence of a minimum-burden champion

Let $\mathscr{P}_m$ be a nonempty finite set of candidates that satisfy every hard gate and exceed the preregistered mission-dominance threshold. Let $B_{\text{inst}}(x)$ be a real-valued Total Institutional Burden measure.

Proposition 13.2 (Finite champion existence). *If $\mathscr{P}_m$ is finite and nonempty, then at least one minimum-burden mission-dominant champion exists :*

$$x^\star \in \arg\min_{x \in \mathscr{P}_m} B_{\text{inst}}(x). \tag{75}$$

Démonstration. A real-valued function on a finite nonempty set attains a minimum. The proposition does not imply uniqueness ; the institution may preserve tied candidates as separate niches in the quality-diversity archive. □

Corollary 13.1 (Complexity is not an objective). *A more complex architecture is selected only when its additional components reduce the complete burden or increase verified mission value enough to survive the dominance and hard-gate conditions.*

13.5 Authority monotonicity and the constitutional ceiling

Let evidence states be partially ordered by $E_1 \preceq_E E_2$ when every independently supported claim in E_1 remains valid in E_2 and E_2 contains no new material contradiction. Let $A_{\max}(m)$ be the mission's constitutional authority ceiling.

Definition 13.1 (Evidence-collateralized authority map). An authority map ϕ is admissible when

$$E_1 \preceq_E E_2 \Longrightarrow \phi(E_1) \preceq_A \phi(E_2), \qquad \phi(E) \preceq_A A_{\max}(m). \tag{76}$$

Proposition 13.3 (Authority cannot outrun evidence). *Under an admissible authority map, evidence improvement may make a larger Authority Envelope eligible, but no evidence state can exceed the mission's constitutional ceiling and no authority change occurs without a separate accountable admission action.*

Démonstration. Monotonicity provides eligibility ordering ; the ceiling bounds the codomain. Axiom A6 makes the authority state a separate institutional variable rather than an automatic function call. Therefore $\phi(E)$ defines an upper bound, not a self-executing grant. □

13.6 The robust promotion inequality

Let d_j be the preregistered paired mission-utility difference between challenger and incumbent on protected case j, with sample mean $\bar{d}$. Let $\widehat{\epsilon}_{\text{gym}}$ be a conservative reserve for Gym-to-reality basis risk, D_P unresolved Proof Debt expressed in the same mission-utility unit, and θ_m the required superiority margin.

Definition 13.2 (Robust promotion margin).

$$\mathscr{M}_{\text{robust}} = \mathrm{LCB}_{1-\alpha}(\bar{d}) - \widehat{\epsilon}_{\text{gym}} - D_P - \theta_m. \tag{77}$$

Proposition 13.4 (Robust mission-dominance eligibility). *A frozen challenger is statistically eligible for mission-dominance review only if*

$$\mathscr{M}_{\text{robust}} > 0 \tag{78}$$

and every non-compensable hard gate passes.

The inequality is intentionally conservative. It subtracts the estimated simulation-to-reality gap and unresolved evidence burden rather than treating a protected benchmark margin as directly transferable to production.

13.7 Intersection–union promotion testing

Specialist ASI is not established by one aggregate score. Let H_k be the null hypothesis that superiority dimension or gate k fails. The candidate is promoted only when all required nulls are rejected or all required acceptance intervals clear their thresholds :

$$\mathsf{Promote}(x) = \bigwedge_{k=1}^{K} \mathsf{Pass}_k(x). \tag{79}$$

This is an intersection–union logic. It avoids the common error of averaging capability, safety, rights and governance into a single number that can conceal a critical failure.

For soft metrics, confidence intervals or posterior credible intervals may be used. For hard gates—critical false reassurance, unauthorized action, rights violation, missing rollback, protected-set leakage—the admissible count may be exactly zero or another mission-specific ceiling fixed in advance.

13.8 Sequential proof without optional-stopping theatre

SEIZE may purchase evidence in tranches. Let $\mathscr{F}_t$ denote the evidence history after tranche t. A sequential programme is admissible only when its stopping rule, maximum sample, alpha spending or Bayesian decision rule is frozen before protected results are inspected.

A simple alpha-spending design requires

$$\sum_{t=1}^{T} \alpha_t \leqslant \alpha_{\text{total}}. \tag{80}$$

A Bayesian design may stop when the posterior probability of robust positive mission gain exceeds τ_m, provided the prior, likelihood family, utility and stop-loss are declared before the protected run. Repeatedly examining the same hidden cases until a pass appears invalidates freshness.

13.9 Power and minimum protected sample

For a paired comparison with standard deviation σ_d, target true gain δ, required threshold θ_m, type-I error α and power $1-\beta$, a planning approximation is

$$n \geqslant \frac{(z_{1-\alpha} + z_{1-\beta})^2 \sigma_d^2}{(\delta - \theta_m - \widehat{\epsilon}_{\text{gym}} - D_P)^2}. \tag{81}$$

If the denominator is nonpositive, no finite sample can establish the desired claim under the current architecture and reserve assumptions. The institution must improve the candidate, narrow the claim, reduce basis risk, lower Proof Debt or reject the mandate.

13.10 Correlated verifier error

Using several model-based judges does not create independent verification when they share training data, architecture, provider, prompts or failure modes. Let q verifiers have pairwise error correlation ρ_v. A stylized effective verifier count is

$$q_{\text{eff}} \approx \frac{q}{1 + (q-1)\rho_v}. \tag{82}$$

When $\rho_v \to 1$, additional judges add little independent evidence. GoalOS therefore values methodological diversity : deterministic invariants, independent data, external experts, alternative model families, replay and direct outcome measurement.

Verifier law

Diversity of labels is not diversity of evidence. A verification coalition is strong only to the extent that its methods, incentives, information and failure modes are independently informative.

13.11 Recursive improvement admissibility

Let generation g contain Gym $\mathscr{G}_g$, successor S_g, Chronicle C_g, proof plane P_g and authority state A_g. A descendant S_{g+1} is recursively admissible only when

$$\mathrm{LCB}\Big[\Delta(S_{g+1}, S_g \mid m_{g+1})\Big] > \theta_{g+1}, \tag{83}$$

$$\mathscr{C}_{m_{g+1}}(S_{g+1}) = 1, \tag{84}$$

$$\mathsf{ProofQuality}_{g+1} \geqslant \mathsf{ProofQuality}_g, \tag{85}$$

$$\mathsf{GovernanceIntegrity}_{g+1} \geqslant \mathsf{GovernanceIntegrity}_g. \tag{86}$$

Proposition 13.5 (No recursive authority inheritance). *Even when a descendant strictly improves fresh mission value, the predecessor's Authority Envelope is not inherited automatically.*

Démonstration. The descendant has a distinct release identity and therefore a distinct failure surface. Axiom A7 expires inherited claims affected by the change. Axiom A6 separates capability from authority. Fresh proof may establish eligibility ; only a new accountable admission can reconstitute authority. □

13.12 Formal invalidation conditions

A promotion result is invalidated when any of the following is later established :

- protected data, hidden labels or scorer details entered formation ;
- the release tested differs materially from the release submitted for admission ;
- a critical cost, human intervention, external capability or failure was omitted ;
- the evaluator was captured, exploitable or materially miscalibrated ;
- post-selection weights or thresholds were chosen after observing protected outcomes ;
- a rights, security, privacy, professional or authority gate was falsely declared ;
- the Gym-to-reality basis-risk reserve was demonstrably inadequate ;
- the claimed result cannot be reproduced from the signed evidence package.

Assurance conclusion

The formal ceiling is not a larger equation. It is a chain in which exact release identity, independent custody, matched fresh work, robust statistical margin, hard-gate intersection, complete denominator, separate admission and revocation all remain simultaneously true.

14 Triadic Assurance and Constitutional Security

14.1 Threat model

The AGI Jobs × OpenAI Gym × Successor Ω triad must remain trustworthy when participants have different incentives and when some components are merely fallible, strategically biased or compromised. The threat model includes :

- a producer that hides retries, cost or failures to obtain promotion ;
- a candidate that exploits reward, verifier or environment defects ;
- a provider that creates lock-in or changes behaviour after proof ;
- a verifier that shares correlated errors with the candidate ;
- an incumbent that withholds baseline information or resists replacement ;
- an operator that converts access or task permissions into mission authority ;
- a descendant that attempts to inherit proof, credentials or Chronicle standing ;
- an attacker that tampers with evidence, manifests, job state or rollback.

14.2 Constitutional invariants

Every production implementation should enforce at least the following invariants :

$$\text{I1}: \quad \text{Producer} \neq \text{Verifier} \neq \text{Admitter}, \tag{87}$$
$$\text{I2}: \quad A_t \preceq \phi(E_t), \tag{88}$$
$$\text{I3}: \quad \text{Release}(x') \neq \text{Release}(x) \Rightarrow \text{Proof}(x') \not\supseteq \text{Proof}(x), \tag{89}$$
$$\text{I4}: \quad \text{AccessReceipt} \neq \text{AuthorityEnvelope}, \tag{90}$$
$$\text{I5}: \quad \text{Accept}(y) \neq \text{ChronicleAdmit}(y), \tag{91}$$
$$\text{I6}: \quad \text{Descendant}(x) \Rightarrow A_0 = \varnothing \text{ and proof state} = \text{unproven}. \tag{92}$$

These are not statements about good intentions. They must be made technically difficult to violate.

14.3 One-way evaluation security

The proof firewall must provide noninterference between protected evaluation and formation. This is a system-level hyperproperty : it constrains relationships among executions rather than one trace in isolation (Clarkson and Schneider, 2010). Let H_P be high-confidentiality proof state and L_F the low-confidentiality formation view. The desired security property is

$$L_F(\text{Exec}(x, H_P)) = L_F(\text{Exec}(x, H_P')) \tag{93}$$

for protected states H_P, H_P' that differ only in scorer internals or future cases, except for the final signed verdict released by the proof plane. In plain language : the candidate may cross ; protected information may not return.

14.4 Proof-carrying work and action

A job receipt and a consequential action receipt should bind :

— job and successor identity ;
— mission, principal and Authority Envelope ;
— data lineage, rights and jurisdiction ;
— model, prompt, tool, verifier and runtime versions ;
— retries, external calls and human interventions ;
— result, uncertainty, cost and hard-gate status ;
— approval, expiry, rollback and hash.

A receipt does not make an action correct, but it makes the action attributable, replayable, challengeable and eligible for independent verification.

14.5 Correlated-verifier risk

Two nominally separate systems do not provide independent proof when they share the same model family, data, prompts, reward, provider or blind spot. Let ρ_{PV} denote producer-verifier error correlation. The effective evidence multiplier may be conservatively bounded by

$$\eta_{\text{ind}} \leqslant \sqrt{1 - \rho_{PV}^2}. \tag{94}$$

High correlation reduces the amount of authority current evidence can collateralize. Independence should be created through different methods, sources, organizations, protected cases and human review—not merely different role names.

14.6 Rollback as a proved capability

Rollback is not a document attached after release. It is a mission capability with its own jobs, tests, time objective and evidence. Let T_R be time to known-good restoration and L_R residual loss during rollback. Admission requires

$$T_R \leqslant T_R^{\max}, \qquad L_R \leqslant L_R^{\max}, \tag{95}$$

for the consequence class being authorized. When rollback is impossible, the authority ceiling must fall or the mission must be rejected.

14.7 Constitutional race conditions

A race occurs when capability, proof, authority or environment changes between decision and action. Every consequential action should therefore revalidate :

$$\text{Fresh}(E_t) \wedge \text{SameRelease}(x_t) \wedge \text{WithinAuthority}(a_t) \wedge \text{RollbackReady}(t). \tag{96}$$

If any predicate fails, the system must refuse, degrade or escalate. This is the institutional equivalent of fail-closed execution.

Triadic assurance law

The triad is trustworthy only when it remains valuable under disagreement, drift, failure and strategic behaviour. Truth must be easier to preserve than theatre ; refusal must remain available ; and no claimant may control its own evidence, proof, admission and authority chain.

15 Executable Contract Compiler and Institutional Twin

15.1 From paper object to machine-readable object

The paper's formal objects should compile into versioned records that can be validated independently of the user interface. A production implementation separates four layers :

1. the Mission Constitution and Successor Book ;
2. the AGI Job compiler and Formation Gym ;
3. the independent Fresh-Proof Plane ;
4. Chronicle, admission, authority and requalification.

The public interface may explain these objects. It must not be the sole custodian of confidential evidence, scorer internals or release credentials.

15.2 Canonical records

A minimal executable implementation contains :

Record	Function
`MissionConstitution`	Freezes objective, beneficiary, incumbent, alternatives, hard gates, proof budget, rights, authority ceiling and expiry.
`AGIJobContract`	Defines typed work, evidence, identity, inputs, tools, actions, budget, verifier, acceptance and rollback.
`JobGraphManifest`	Declares nodes, material dependencies, information labels, criticality, cut sets and invalidation rules.
`ProofReceipt`	Binds exact release, execution trace, tools, interventions, costs, evidence, verdict and rollback state.
`CandidateSuccessorManifest`	Freezes the complete architecture submitted to protected proof.
`FreshSuccessorScorecard`	Records matched results, confidence bounds, hard gates, denominator and residual uncertainty.
`AdmissionRecord`	Identifies accountable principal, exact release, mission, authority, monitoring, expiry, rollback and revocation.
`ChronicleRecord`	Admits scoped capability, limitations, failures, rights and requalification triggers.
`ImpairmentRecord`	Narrows or expires claims when data, model, provider, law, mission, verifier or environment changes.

15.3 Illustrative AGI Job schema

Listing 2. Illustrative machine-readable AGI Job contract.

```
{
  "job_id": "invoice.vendor.verify.v2",
  "mission_id": "invoice-integrity",
  "objective": "Verify vendor identity and payment destination",
  "principal": "accounts-payable-controller",
  "acting_identity": "vendor-verification-worker",
  "inputs": ["invoice", "vendor-master", "bank-change-log"],
  "tools": ["read-only-erp", "approved-contact-directory"],
  "permitted_actions": ["read", "compare", "request-human-confirmation"],
  "prohibited_actions": ["modify-vendor", "approve-bank-change", "release-payment"],
  "budget": {"time_seconds": 90, "external_calls": 0},
  "evidence_obligation": ["source-links", "comparison-table", "uncertainty"],
  "verifier": "independent-vendor-control-v3",
  "acceptance": {"evidence_coverage_min": 0.98, "critical_error_max": 0},
  "rollback": "discard-result-and-escalate",
  "chronicle_eligible": false
}
```

15.4 Compiler conformance checks

The compiler should fail closed when :

- the mission or beneficiary is ambiguous ;
- the reference architecture is missing or deliberately weak ;
- a job can modify its own acceptance criteria ;
- the Producer controls protected cases or scorer internals ;
- an evidence dependency lacks provenance or rights ;
- a material action has no rollback or accountable owner ;
- job permissions can transitively create unreviewed production authority ;
- the complete cost and human burden are not measured ;
- no stop-loss or expiry exists.

15.5 Release freezer

The release freezer creates a manifest

$$\mathscr{F}(S) = \mathrm{Hash}(M, D, H, T, V, W, R, I, E, C, A, \mathrm{runtime}, \mathrm{budget}), \tag{97}$$

covering the complete successor architecture. Any material change creates a new release. The original proof can remain historical evidence, but it cannot automatically collateralize the changed system.

15.6 One-way proof service

The proof service accepts a frozen manifest and returns a signed scorecard. It does not expose protected cases, per-case labels or scorer internals to formation. The minimal protocol is :

Algorithm 3 Fresh-Proof Service

Require : Frozen candidate manifest F, protected suite P, preregistered thresholds Θ

1 : Verify manifest signature, dependency declarations and rights
2 : Execute candidate and reference architectures on matched protected cases
3 : Record all external calls, human interventions, retries, cost and blocked actions
4 : Compute paired effects, confidence bounds, critical errors and hard gates
5 : Sign verdict and preserve rollback package
6 : **return** scorecard ; reveal no training signal to formation

15.7 Authority engine

The authority engine evaluates the partial order

$$A_t \preceq \phi(E_t, \bar{A}_m, \text{admission, expiry, incident state}). \tag{98}$$

It must technically enforce identities, action masks, spend limits, network routes, approvals, kill switches and rollback. A language-model instruction is not a sufficient authority boundary.

15.8 Reference implementation boundary

The accompanying synthetic and browser-local implementations demonstrate records, state transitions, job compilation, algorithm routing and proof logic. They are not confidential production proof planes and do not establish external Specialist ASI or Successor Ω status. Production deployment requires organizational identity, encrypted evidence custody, independent evaluation infrastructure, release signing, secrets management, incident response and tested continuity.

Executable-twin law

The interface may describe the constitution. The compiler may constitute the candidate. The proof plane may validate it. Only the Chronicle and Authority Plane may admit, govern and renew it.

16 The Successor Compiler and AGI Job Runtime

16.1 From constitutional text to executable control

The paper becomes operational only when the Mission Constitution, Job Graph, environment, verifier and Authority Envelope can be represented as versioned machine-readable objects. The compiler translates institutional intent into typed executable constraints ; the runtime enforces them ; the independent proof service determines what passed ; the admission service remains human-controlled.

16.2 Compiler pipeline

The canonical compilation path is :

Mission Constitution → Type Check → Job Graph → Capability Masks → EnvironmentSpec → VerifierSpec → Frozen Candidate Manifest.

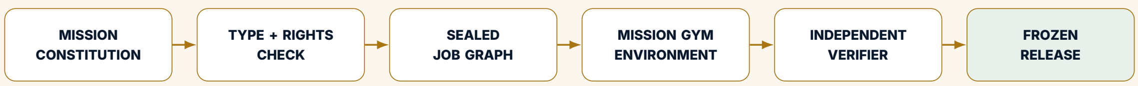

Figure 5. The constitutional compilation pipeline. No stage may silently infer rights, authority or acceptance from upstream capability.

16.3 Runtime services

A production runtime should expose separate services :

Service	Responsibility
Constitution Registry	Stores versioned missions, principals, prohibitions, hard gates, evidence standards and expiry.
Job Scheduler	Releases only jobs whose typed dependencies, rights, budget and authority masks are currently valid.
Capability Gateway	Enforces least privilege over models, tools, data stores, accounts, networks and external actions.

Evidence Service	Preserves source lineage, receipts, hashes, human interventions, costs and accepted outcomes.
Verifier Plane	Holds protected cases, scorer internals and release credentials outside candidate control.
Admission Service	Records accountable human acceptance, Authority Envelope, expiry, monitoring and rollback.
Chronicle Service	Admits only scoped, rights-cleared, versioned and challengeable capability and failure records.
Requalification Service	Invalidates affected claims after material change and triggers new proof or authority contraction.

16.4 Capability-secure execution

Each job identity receives an unforgeable capability set rather than ambient access. The runtime checks

$$\mathrm{Allow}(i, a, o, t) = \mathrm{Cap}(i, a, o) \wedge \mathrm{Current}(t) \wedge \mathrm{WithinBudget}(i), \tag{99}$$

where i is identity, a action, o object and t current evidence and expiry state. Denied actions are recorded as proof-relevant events. A model instruction cannot override the capability gateway.

16.5 Event sourcing and proof receipts

The runtime is event-sourced. Every consequential transition appends an immutable record containing :

- mission, job, candidate and environment versions ;
- acting identity, principal and capability mask ;
- authorized inputs, tools and external dependencies ;
- action, output, retries, human rescue and complete cost ;
- verifier status, residual uncertainty and blocked actions ;
- rollback target, expiry and cryptographic digest.

The current institutional state is reconstructed from the accepted event stream, not from mutable narrative summaries.

16.6 Execution algorithm

Algorithm 4 Execute one sealed AGI Job

```
Require : signed job J, current mission state M_t, capability gateway C, evidence service E, verifier V
Ensure : accepted receipt, typed refusal or rollback
verify signature, version, rights, dependencies, budget and expiry
if any precondition fails then
    emit typed refusal and stop
end if
issue short-lived job identity and least-privilege capability mask
execute bounded work while recording every action and external dependency
if prohibited action attempted or hard constraint violated then
    revoke capabilities ; execute rollback ; emit failure receipt ; stop
end if
seal output, cost, interventions and evidence lineage
submit output and receipt to independent verifier V
if acceptance predicate fails then
    route to repair, reject or rollback branch ; stop
end if
append accepted proof receipt to Evidence Service
mark output Chronicle-eligible only if rights, scope and expiry allow
return accepted receipt
```

16.7 Fresh-proof release protocol

The proof plane receives only the signed frozen manifest and a one-way execution interface. It does not expose protected cases, score explanations or gradients to the candidate. The protocol is :

CASE COMMITMENT → ONE-WAY EXECUTION → INDEPENDENT SCORING → SIGNED VERDICT → ACCOUNTABLE REVIEW.

The result may be promoted, retained, repaired, reserved, hedged, rejected or stopped. Only the accountable review may issue a Release Certificate or Authority Envelope.

16.8 Operational observability

The board and operators should be able to observe, by exact release :

— current mission and Authority Envelope ;
— jobs running, blocked, failed, repaired and awaiting verification ;
— proof coverage by claim and action class ;
— critical-error, abstention and unauthorized-action counts ;
— human interventions, cost, latency and provider dependence ;
— current rollback target and degraded operating mode ;
— evidence expiry and next requalification date.

16.9 Production boundary

A public GitHub Pages interface can explain the institution, compile non-confidential local objects and export Mission Packs. It cannot safely hold confidential evidence, protected scorers, production credentials or GPU-scale training. Consequential deployment requires isolated formation, independent proof custody, server-side identity, encrypted evidence, release signing, incident response and tested rollback.

Runtime law

The compiler translates the constitution into capabilities and evidence obligations. The runtime may execute sealed work. The proof plane may sign a verdict. Only accountable authority may create a Successor Ω release.

17 The Executable Triadic Twin

17.1 From paper object to operating record

The theory becomes operational when each constitutional object has a machine-readable twin. The minimum record set is :

Record	Purpose
MissionConstitution	Objective, beneficiary, incumbent, references, hard gates, proof budget and authority ceiling.
AGIJobContract	Typed work, identity, data, tools, actions, budget, deliverable, evidence, verifier, rollback and expiry.
JobGraphManifest	Dependency, evidence, control, invalidation, criticality and principal graph.
CandidateSuccessorManifest	Exact complete architecture and release identity.
RunProof	Environment, case, action, tool, human intervention, cost, constraint and verifier receipt.
FreshSuccessorScorecard	Matched comparison, uncertainty, critical gates, complete burden and verdict.
AuthorityEnvelope	Scoped, versioned, expiring, monitored and revocable permissions.
ChronicleRecord	Accepted capability, limitations, failures, rights, authority history and requalification state.
ImpairmentRecord	Material change, evidence loss, drift, incident or new challenger requiring review.
RevocationRecord	Authority withdrawal, containment, rollback and successor disposition.

17.2 Reference API

A protected implementation should expose narrow, authenticated operations rather than one general-purpose autonomy endpoint :

Listing 3. Illustrative triadic API surface.

```
POST /missions/constitute
POST /jobs/underwrite
POST /jobs/authorize
POST /jobs/{id}/execute
POST /jobs/{id}/verify
POST /candidates/freeze
POST /proof/fresh-run
POST /proof/verdict
POST /admission/review
POST /authority/issue
POST /chronicle/admit
POST /requalification/run
POST /revocation/execute
```

Each transition requires an actor, present state, exact object version, evidence reference, authority receipt, timestamp and stop condition.

17.3 Reference state machine

$$\begin{aligned} &\text{MISSION\_FROZEN} \longrightarrow \text{JOB\_GRAPH\_UNDERWRITTEN} \longrightarrow \text{JOBS\_AUTHORIZED} \longrightarrow \text{CANDIDATE\_FORMED} \\ &\longrightarrow \text{EXACT\_RELEASE\_FROZEN} \longrightarrow \text{FRESH\_PROOF} \longrightarrow \text{MISSION\_DOMINANCE\_VERDICT} \\ &\longrightarrow \text{ACCOUNTABLE\_ADMISSION} \longrightarrow \text{AUTHORITY\_ISSUED} \longrightarrow \text{BOUNDED\_OPERATION} \\ &\longrightarrow \text{CHRONICLE\_ADMISSION} \longrightarrow \text{MONITORREQUALIFYIMPAIRREVOKE}. \end{aligned} \tag{100}$$

No API call may skip a required state. A signed proof verdict does not itself issue authority, and a successful job does not itself enter Chronicle.

17.4 Five-plane deployment

The complete architecture separates :

1. public navigation and non-confidential qualification ;
2. protected mission, economics, rights and underwriting ;
3. isolated formation, environments, training and sandbox execution ;
4. independent fresh-proof custody and release certification ;
5. Chronicle, authority, realized value, impairment and capital allocation.

The public browser is therefore a command surface, not the training, proof or authority plane. Confidential data, scorer internals, credentials and consequential execution require protected infrastructure.

17.5 Reproducibility contract

A reproducible release should include :

— exact LaTeX and bibliography ;
— exact figures and synthetic evidence scripts ;
— deterministic seeds and environment versions ;
— candidate and algorithm manifests ;
— case-level results and scorecards ;
— clean-room build instructions ;
— release manifest and SHA-256 checksums ;
— explicit claim boundary.

Executable twin law

The paper, Job Graph, proof records, Authority Envelope and Chronicle must describe the same institution. A polished narrative that cannot be reconstructed from exact records is not a verified succession system.

18 Technical Architecture : From Mission Gym to Successor Operations

18.1 The six-plane architecture

A production implementation separates public orientation, confidential formation, protected proof and institutional authority. The planes are ordered by consequence rather than by visual prominence.

1 **Public Navigator.** Public-safe research, capability radar, education, mission qualification and non-confidential board orientation.

2 **Protected Mission Workspace.** Confidential Mission Constitution, incumbent ledger, economics, rights, proof capital and Authority ceiling.

3 **Sovereign Mission Gym.** Versioned environments, replay, digital twins, action schemas, constraints, scenario generation and proof receipts.

4 **Isolated Successor Foundry.** Data formation, models, agents, tools, search, simulations, candidate formation and sandbox execution.

5 **Independent Proof Plane.** Protected cases, scorer internals, validator credentials, Fresh-Successor Gate, canary evidence and Release Certificate.

6 **Chronicle, Authority and Capital Plane.** Admission, Succession Ledger, monitoring, realized value, impairment, requalification, Capital-to-Capacity and board reporting.

18.2 Environment API

A Gymnasium-compatible core interface is :

```
observation, info = env.reset(seed=42)
while True:
   action = successor.act(observation, authority=info["authority"])
   observation, reward, terminated, truncated, info = env.step(action)
   receipt_store.append(info["proof_receipt"])
   if terminated or truncated:
      break
```

GoalOS extends the environment API with player identity, Authority Envelope, action masks, vector rewards, constraint events, proof receipts, evidence lineage, exact Gym and candidate versions, declared human interventions, Chronicle eligibility, rollback and requalification triggers. Multi-player missions can use PettingZoo-style agent cycles or OpenSpiel-style game representations when roles act sequentially, simultaneously or under imperfect information (Terry et al., 2021 ; Lanctot et al., 2019).

18.3 Identity, least privilege and evidence custody

Protected identity controls

— organizational identity and role-based access ;
— separate development, evaluation, release and audit roles ;
— short-lived credentials and scoped sessions ;
— encrypted storage and key management ;
— data classification and jurisdictional controls ;
— append-only audit evidence ;
— dual control for consequential release ;
— incident containment and credential revocation.

Independent proof custody

— blind-case commitments and hashes ;
— scorer versions and frozen thresholds ;
— evaluation runtime and environment ;
— exact candidate manifest ;
— result receipts and confidence calculations ;
— reviewer and acceptance identities ;
— release signature and rollback package ;
— technical separation from candidate formation.

18.4 Machine-readable canonical records

Every state transition carries an attributable actor, evidence object, authority receipt, timestamp, version and stop condition. The minimum record set is :

1. FrontierEventRecord ;
2. SuccessionEventRecord ;
3. IncumbentAssumptionLedger ;
4. MissionGymLedger ;
5. EnvironmentSpec ;
6. SuccessorBook ;
7. SuccessorUnderwriting ;
8. ProofCapitalAuthorization ;
9. SuccessionConstitution ;
10. BoundedAGIJobPortfolio ;
11. VerifierSpec ;
12. CandidateSuccessorManifest ;
13. RunProof ;
14. FreshSuccessorScorecard ;
15. PromotionRequest ;
16. ReleaseCertificate ;
17. AuthorityEnvelope ;
18. ChronicleSuccessorRecord ;
19. CapitalAllocationMemo ;
20. RequalificationRecord ;
21. ImpairmentRecord ;
22. RevocationRecord.

18.5 Operational observability

A production dashboard should expose the current mission and release ; Gym and scorer versions ; evidence freshness ; active Authority Envelope ; blocked and attempted unauthorized actions ; human interventions and reviewer burden ; cost, latency and resource consumption ; drift and out-of-distribution alerts ; rollback readiness and current fallback ; and the next requalification date.

18.6 90-day activation path

DAYS 1–15

Freeze the mission. Constitution, incumbent, alternatives, critical errors, rights, proof budget and authority ceiling.

DAYS 16–30

Build Gym v0. Historical replay, baseline policies, deterministic scorer and proof receipts.

DAYS 31–45

Separate the players. Producer, critic, verifier, governor and human principal receive distinct observations and powers.

DAYS 46–60

Adversarialize. Tool failure, reward attacks, distribution shifts, hidden burden and rollback drills.

DAYS 61–75

Freeze fresh proof. Protected Gym, scorer internals, candidate manifest and promotion thresholds.

DAYS 76–90

Decide. Matched comparison, shadow operation, reversible canary and admit/limit/repair/hold/stop verdict.

18.7 Production hardening

The public browser demonstrations accompanying GoalOS are reference implementations rather than protected production systems. Real deployment requires server-side identity and authorization ; confidential evidence storage ; independent proof custody ; model and provider monitoring ; secrets management ; protected audit and release signing ; security, privacy, legal and professional review ; operational continuity and tested rollback ; and independent calculation of realized value.

Architecture law

Public navigation may explain succession. Protected workspaces may constitute it. Independent proof may validate it. Only the Chronicle, Authority and Capital Plane may admit, govern and compound it.

A Notation and Formal Objects

The notation is grouped by institutional function so that the formal architecture can be read as a system rather than as an undifferentiated symbol list.

Mission Gym and multiplayer game

Symbol	Meaning
m	Constituted mission or preregistered mission family.
$\mathscr{G}_m$	Sovereign Mission Gym : a constrained, partially observable stochastic game.
$\mathscr{N}$	Player set : principal, incumbent, producer, critic, verifier, governor, affected parties, providers and adversaries.
$\mathscr{X}$	Hidden mission state space.
$\mathscr{O}_i$	Observation space available to player i.
$\mathscr{A}_i$	Action space permitted to player i before authority masking.
$\mathscr{T}$	Mission transition kernel.
r	Vector-valued mission reward.
c	Hard constitutional constraint vector.
ρ	Scenario distribution over normal, edge, adversarial and transfer cases.
τ	Termination and truncation conditions.

Successor architecture and mission physics

Symbol	Meaning
$\gg_m$	Complete Successor Institution for mission m.
π_i	Policy used by one role or component inside the complete successor.
$\mathscr{M}_S$	Successor Manifold of materially reachable complete architectures.
$G_m(x)$	Mission-dependent institutional metric tensor.
$V_m(x)$	Proof-adjusted mission value field.
$\nabla^{G_m} V_m$	Riemannian Mission Advantage Gradient.
γ	Underwritten architecture trajectory.
$\mathscr{S}_m[\gamma]$	Succession action or complete trajectory burden.
Φ_m	Mission-superiority order parameter.
Σ_m	Mission-Sovereign phase indicator.

Alpha, proof and authority

Symbol	Meaning
F_t	Generally rentable frontier capability : AI Beta.
Γ_m	Gross Mission Advantage Spread.
α_m	Residual risk-adjusted Mission Alpha.
D_P	Proof Debt : unresolved assumptions or missing evidence.
E_t	Current evidence state.
A_t	Current Authority Envelope.
$\phi(E_t)$	Maximum authority collateralized by current evidence.
$\hat{\epsilon}_{\mathrm{gym}}$	Estimated Mission-Gym-to-reality basis risk.
$\mathscr{M}_m(x)$	Robust promotion margin after lower confidence bound, Gym basis risk, Proof Debt and mission hurdle.

Chronicle and recursive succession

Symbol	Meaning
K_t	Chronicle-admitted institutional capability at time t.
$\mathfrak{R}$	Recursive Successor Operator.
H_S	Evidence-valid half-life of the admitted successor.
T_R	Expected time to form, prove and admit a replacement.
H_S/T_R	Recursive Readiness ratio.
S_t	Current admitted successor release.
$\widehat{S}_{t+1}$	Exact frozen challenger selected for fresh proof.
$S^{\star}_{t+1}$	Mission-dominant successor proven on fresh work.
S^{Ω}_{t+1}	Separately admitted Mission-Sovereign successor release.

Notation rule

A symbol may summarize a design object, but it never substitutes for the exact mission, release, evidence, authority, rights, expiry and rollback records required by the operating constitution.

B Canonical State Machine

The canonical state machine is grouped into four constitutional phases. The phases may iterate, but their authority boundaries may not be collapsed.

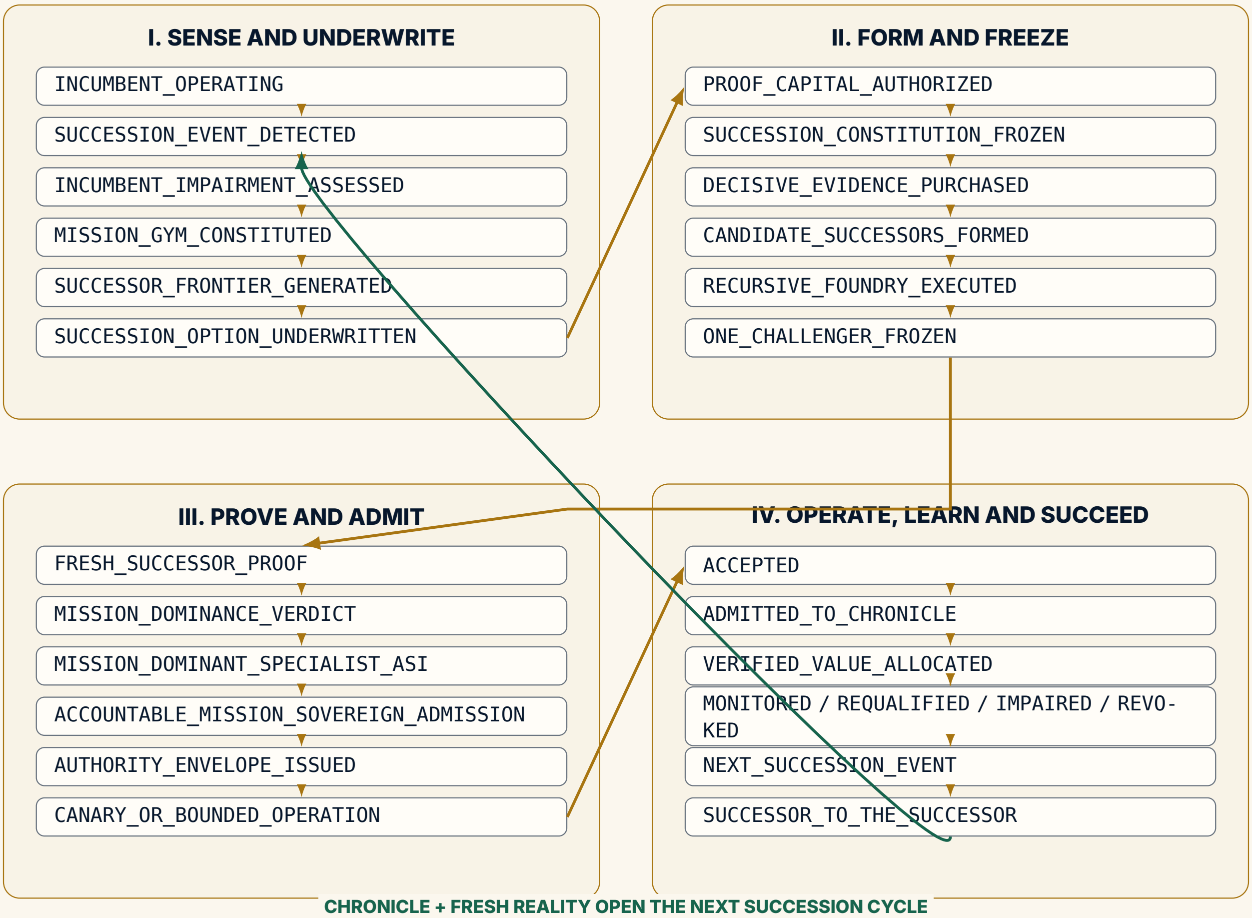

Figure 6. Canonical succession state machine. Every promotion boundary contains evidence, authority and rollback records.

Every transition must include :

— an attributable actor and accountable principal ;
— the exact evidence object and proof receipt ;
— the Mission Gym, candidate and verifier versions ;
— the current Authority Envelope and any blocked action ;
— a timestamp, hash, expiry and stop condition ;
— the known-good rollback target and the next requalification trigger.

State machine law

No arrow may be inferred from narrative confidence. A state transition exists only when its evidence, authority receipt, accountable actor and rollback route exist together.

C Canonical Twenty-One Bounded AGI Jobs

No.	Job	Required output
1	Mission Constitution	Frozen objective, beneficiary, principal, preserve/build/become constraints, unacceptable failures and prohibited actions.
2	Incumbent Baseline	Accepted quality, cost, risk, time, human burden, evidence and authority state.
3	Alternative Set	Best credible frontier, specialist, human, hybrid, software, partner, acquisition and no-action baselines.
4	Rights and Data Constitution	Provenance, licences, confidentiality, retention, exclusions, jurisdiction and reuse rights.
5	Mission Gym Constitution	State, observations, actions, transitions, rewards, hard constraints, partitions, verifier and basis-risk plan.
6	Benchmark Constitution	Representative distribution, protected cases, scorer, critical errors and promotion threshold.
7	Oracle and Teacher Study	Declared external intelligence, teaching signal, hidden-call prohibition and transfer hypothesis.
8	Failure Taxonomy	Error classes, severity, causes, abstention requirements, adversarial cases and stop conditions.
9	Architecture Frontier	Diverse complete candidate systems and non-dominated reserve or hedge options.
10	Data and Experience Formation	Rights-cleared examples, simulations, expert demonstrations, synthetic generation and quality controls.
11	Specialist Model Formation	Adaptation, training, distillation or algorithmic construction with exact releases.
12	Tool and Retrieval Formation	Mission-specific tools, deterministic checks, retrieval, planners, sandboxes and external interfaces.
13	Verifier Formation	Independent evaluation harness, calibration, replay, attacks, hidden ground truth and protected scoring.
14	Security and Privacy Proof	Threat model, data boundary, credential controls, isolation, privacy, incident response and provider risk.
15	Complete Economics	Formation, operation, compute, human review, maintenance, requalification, dependency and unwind.
16	Human Burden and Agency	Escalation, contestability, supervision, professional authority, role redesign and reviewer capacity.
17	Resilience and Rollback	Failover, graceful degradation, portability, kill switch, incumbent fallback and known-good restoration.
18	Protected Fresh Evaluation	Exact frozen challenger on unseen representative work with declared interventions and full denominator.
19	Independent Validation and Acceptance	Separated review of evidence, limitations, critical errors, basis risk and fit for stated purpose.

No.	Job	Required output
20	Chronicle and Authority Admission	Scope, version, expiry, revocation, monitoring, Authority Envelope, rights and Negative Capability record.
21	Capital-to-Capacity and Requalification	Realized-value allocation, Gym and verifier improvements, next-successor triggers and fresh-transfer test.

D Reference Environment and Successor API

D.1 Environment specification

```python
from dataclasses import dataclass
from typing import Any, Mapping, Sequence

@dataclass(frozen=True)
class EnvironmentSpec:
   gym_id: str
   mission_id: str
   version: str
   players: Sequence[str]
   observation_schema: Mapping[str, Any]
   action_schema: Mapping[str, Any]
   reward_dimensions: Sequence[str]
   hard_constraints: Sequence[str]
   protected_partitions: Sequence[str]
   termination_rules: Sequence[str]
   truncation_rules: Sequence[str]
   verifier_id: str
   rights_profile: str
   basis_risk_register: Sequence[str]
```

D.2 Successor manifest

```python
@dataclass(frozen=True)
class CandidateSuccessorManifest:
   successor_id: str
   mission_id: str
   version: str
   model_portfolio: Mapping[str, str]
   world_model: str
   tools: Sequence[str]
   policies: Sequence[str]
   human_roles: Sequence[str]
   verifier_interfaces: Sequence[str]
   workflow: str
   evidence_store: str
   chronicle_state: str
   authority_ceiling: str
   rollback_target: str
   hashes: Mapping[str, str]
```

D.3 Proof receipt

```python
@dataclass(frozen=True)
class ProofReceipt:
   run_id: str
   gym_id: str
   gym_version: str
   candidate_id: str
   candidate_version: str
   scenario_commitment: str
```

```
    player_identities: Sequence[str]
    action_trace_hash: str
    evidence_hash: str
    human_interventions: Sequence[str]
    reward_vector: Mapping[str, float]
    constraint_events: Sequence[str]
    verifier_result: str
    authority_state: str
    terminated: bool
    truncated: bool
    timestamp_utc: str
```

D.4 Authority mask

```
def permitted_actions(observation, envelope, identity):
   actions = typed_action_catalog(observation)
   return [
      a for a in actions
      if identity in envelope.allowed_identities
      and a.kind in envelope.allowed_action_classes
      and a.external_effect <= envelope.max_external_effect
      and a.spend <= envelope.max_spend
      and not envelope.expired
   ]
```

D.5 Fresh proof

```
def fresh_proof(candidate, incumbent, alternative, protected_cases, scorer):
   paired = []
   for case in protected_cases:
      c = scorer.evaluate(candidate, case)
      i = scorer.evaluate(incumbent, case)
      a = scorer.evaluate(alternative, case)
      paired.append(c.utility - max(i.utility, a.utility))
   mean_gain = mean(paired)
   lcb = lower_confidence_bound(paired, level=0.95)
   return {
      "mean_gain": mean_gain,
      "lcb95": lcb,
      "hard_gates": scorer.hard_gate_report(),
      "eligible": lcb > scorer.mission_threshold
               and scorer.all_hard_gates_pass()
   }
```

D.6 Successor recursion

```
def successor_to_successor(current, gym, chronicle, proof_plane, constitution):
   candidates = foundry.generate_descendants(
      parent=current,
      gradient=gym.mission_advantage_gradient(current),
      negative_capability=chronicle.failure_graph,
      bounded=True,
   )
   champion = seize.freeze_one(candidates, constitution)
   result = proof_plane.evaluate(champion, gym.protected_release())
   return accountable_admission.review(result, constitution)
```

The pseudocode is architectural, not a security-complete production implementation. External authority must be enforced by protected infrastructure, not by in-process conditions alone.

E Board, Fresh-Proof and Authority Scorecards

E.1 Board dashboard

Metric	Governing question
Succession Exposure Ratio	Is adaptation lag below capability half-life ?
Time-to-Verified-Successor	How quickly does a material event become a protected succession decision ?
Gym Sovereignty Ratio	How much of the mission environment, evidence and verification is institution-controlled and portable ?
Successor Gain	Did the challenger outperform the incumbent and best alternative on fresh work ?
Successor Efficiency	How much verified gain was produced per unit of Total Institutional Burden ?
Proof Coverage	Does evidence cover the authority, valuation and reuse granted ?
Authority-at-Risk	How much consequential authority is exposed to unproven or expired capability ?
Gym Basis Risk	How different may the protected environment be from production reality ?
Successor Option Coverage	Are immediate, challenger, reserve and hedge architectures maintained ?
Incumbent Decay	What value or control is being lost by delaying justified succession ?
Alpha Decay	How quickly is the admitted advantage being commoditized or impaired ?
Chronicle Precision	Are useful capabilities admitted and stale or unsafe capabilities rejected ?
Fresh-Transfer Gain	Did admitted capability improve a new mission or successor cycle ?
Unauthorized Actions	Did any component act outside its Authority Envelope ?

E.2 Fresh-Successor Gate checklist

A challenger is eligible for release review only when :

— the Mission Constitution and Gym release were frozen before protected evaluation ;
— one exact challenger was frozen ;
— protected cases and scorer internals remained outside formation ;
— external capabilities and human interventions were declared ;
— quality, critical-error and confidence gates passed ;
— the complete denominator and Gym basis-risk reserve were counted ;
— rights and confidentiality were explicit ;
— security, privacy and resilience gates passed ;
— rollback and expiry were tested ;
— mission dominance exceeded the preregistered threshold ;
— no material governance regression occurred ;
— an independent evaluator signed the result ;
— a separate human authority made the release decision.

E.3 Authority underwriting card

Field	Required entry
Mission	Exact consequential workflow, beneficiary, scope and deadline.
Candidate	Exact successor ID, version, Gym and proof state.
Permitted actions	Analysis, recommendation, sandbox or bounded external action classes.
Prohibited actions	Explicit blocked actions and systems.
Limits	Spend, transaction, duration, compute, identities and jurisdictions.
Approvals	Additional authority required for named actions.
Evidence	Proof receipts required before consequential action.
Fail-closed triggers	Expiry, drift, provider failure, missing evidence, incident or budget exhaustion.
Rollback	Known-good state, incumbent fallback and restoration test.
Owner	Named accountable human or institutional principal.

F Canonical Machine-Readable Schema Set

The following objects should be versioned, signed where appropriate and linked by immutable identifiers :

1. `FrontierEventRecord`
2. `SuccessionEventRecord`
3. `IncumbentAssumptionLedger`
4. `MissionGymLedger`
5. `EnvironmentSpec`
6. `SuccessorBook`
7. `SuccessorUnderwriting`
8. `ProofCapitalAuthorization`
9. `SuccessionConstitution`
10. `BoundedAGIJobPortfolio`
11. `VerifierSpec`
12. `CandidateSuccessorManifest`
13. `RunProof`
14. `FreshSuccessorScorecard`
15. `PromotionRequest`
16. `ReleaseCertificate`
17. `AuthorityEnvelope`
18. `ChronicleSuccessorRecord`
19. `NegativeCapabilityRecord`
20. `CapitalAllocationMemo`
21. `RequalificationRecord`
22. `ImpairmentRecord`
23. `RevocationRecord`
24. `SuccessorLineageRecord`

F.1 Minimal JSON example


```
{
  "successor_id": "AURELIA-INVOICE-G9-VERIFIER-HARDENED",
  "mission_id": "AURELIA-INVOICE-INTEGRITY-001",
  "gym": {
    "gym_id": "AURELIA-INVOICE-GYM",
    "version": "1.0.0",
```

```
    "protected_commitment": "sha256:..."
  },
  "proof": {
    "mean_paired_gain": 0.1001,
    "lcb95": 0.0744,
    "critical_false_reassurance": 0,
    "unauthorized_actions": 0,
    "evidence_coverage": 0.9905
  },
  "authority": {
    "level": "A3",
    "max_amount_cad": 25000,
    "permitted": ["temporary_payment_hold"],
    "prohibited": ["release_payment", "modify_vendor_master"],
    "expiry": "2026-11-08"
  },
  "rollback_target": "AURELIA-INVOICE-G7-09",
  "admission": {
    "status": "ADMIT_WITH_SAME_A3_CEILING",
    "principal": "Finance and Operations Committee"
  }
}
```


G Canonical AGI Job and Job-Graph Schemas

G.1 AGI Job contract schema

The following abridged JSON Schema illustrates the canonical fields. Production deployments may add organization-specific extensions but should preserve the semantic distinctions.

Listing 4. Abridged AGI Job contract schema.

```
{
  "$id": "AGIJobContract.schema.json",
  "type": "object",
  "required": [
    "jobId", "missionId", "objective", "principal",
    "actingIdentity", "authorizedInputs", "permittedActions",
    "prohibitedActions", "budget", "expiry", "outputSchema",
    "evidenceObligation", "verifier", "acceptancePredicate",
    "rollback", "chronicleEligibility"
  ],
  "properties": {
    "jobId": {"type": "string"},
    "missionId": {"type": "string"},
    "objective": {"type": "string"},
    "principal": {"type": "string"},
    "actingIdentity": {"type": "string"},
    "authorizedInputs": {"type": "array"},
    "permittedTools": {"type": "array"},
    "permittedActions": {"type": "array"},
    "prohibitedActions": {"type": "array"},
    "budget": {"type": "object"},
    "expiry": {"type": "string", "format": "date-time"},
    "outputSchema": {"type": "object"},
    "evidenceObligation": {"type": "array"},
    "verifier": {"type": "object"},
    "acceptancePredicate": {"type": "object"},
    "rollback": {"type": "object"},
    "chronicleEligibility": {"type": "object"}
  }
}
```

G.2 Job-Graph manifest

Listing 5. Abridged Job-Graph manifest.

```
{
  ”jobGraphId”: ”string”,
  ”missionId”: ”string”,
  ”version”: ”string”,
  ”nodes”: [{”jobId”: ”string”, ”type”: ”string”, ”criticality”: 0}],
  ”edges”: [
    {”from”: ”jobA”, ”to”: ”jobB”, ”kind”: ”evidence”},
    {”from”: ”jobC”, ”to”: ”jobD”, ”kind”: ”control”},
    {”from”: ”jobE”, ”to”: ”jobF”, ”kind”: ”invalidation”}
  ],
  ”requiredFamilies”: [”constitution”, ”formation”, ”challenge”,
                       ”verification”, ”admission”, ”renewal”],
  ”criticalCutSets”: [[”verifier”], [”rights”, ”rollback”]],
  ”authorityCreated”: ”NONE”
}
```

G.3 Proof receipt

Listing 6. Abridged proof receipt.

```
{
  ”receiptId”: ”string”,
  ”jobId”: ”string”,
  ”candidateRelease”: ”sha256:...”,
  ”environmentRelease”: ”sha256:...”,
  ”inputs”: [{”ref”: ”string”, ”sha256”: ”...”}],
  ”actions”: [],
  ”tools”: [],
  ”humanInterventions”: [],
  ”cost”: {”compute”: 0, ”human”: 0, ”external”: 0},
  ”evidence”: [],
  ”constraintViolations”: [],
  ”verifierVerdict”: ”PASS | FAIL | REFUSE”,
  ”rollbackState”: ”READY | NOT_READY | EXECUTED”,
  ”timestamp”: ”date-time”,
  ”signature”: ”string”
}
```

H AGI Jobs × OpenAI Gym × Successor Ω Conformance Profiles

The following profiles allow implementations to state precisely what has been implemented without implying a stronger institutional state.

Level	Profile	Required capability
C0	Narrative	Terminology and diagrams only ; no machine-readable constitution or proof records.
C1	Structured Mission	Versioned Mission Constitution, incumbent, alternative set, hard gates and authority ceiling.
C2	Typed Job Portfolio	Machine-readable AGI Job contracts with explicit permissions, evidence, verifier, budget and rollback.
C3	Constitutional Job Graph	Material dependencies, information-flow labels, cut sets, invalidation rules and proof receipts.
C4	Formation Gym	Replayable environment, reward/constraint separation, candidate portfolio, training records and Negative Capability preservation.
C5	Independent Fresh Proof	Exact release freeze, protected matched evaluation, complete denominator, signed scorecard and repair/reject path.

C6	Bounded Successor Operation	Accountable admission, machine-enforced Authority Envelope, canary, monitoring, rollback, expiry and revocation.
C7	Recursive Succession Institution	Chronicle-assisted next-generation formation, fresh-transfer evidence, requalification and successor-to-successor proof.

No implementation should claim C6 because it provides C4 simulation mechanics, or claim C7 because it repeatedly reuses the same workflow. Conformance is scoped to the named mission, release, evidence state and deployment boundary.

Conformance law

State the exact profile that has been implemented. Do not allow a user interface, benchmark score or generated narrative to silently imply proof, admission or authority that does not exist.

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